

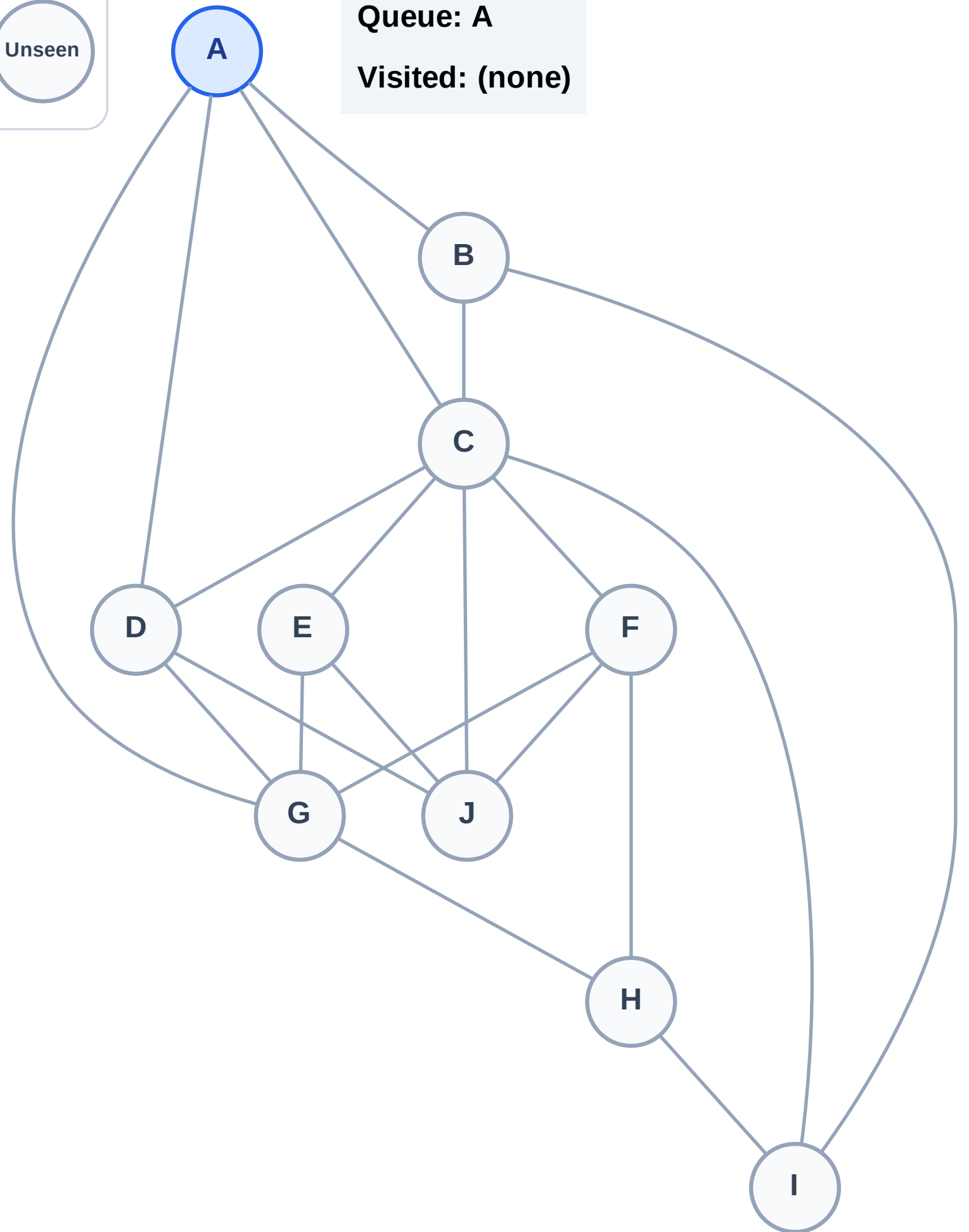
BFS Traversal

Step 1: Start at A

Legend



Queue: A
Visited: (none)



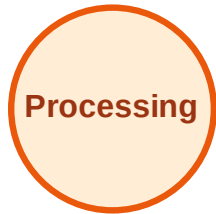
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



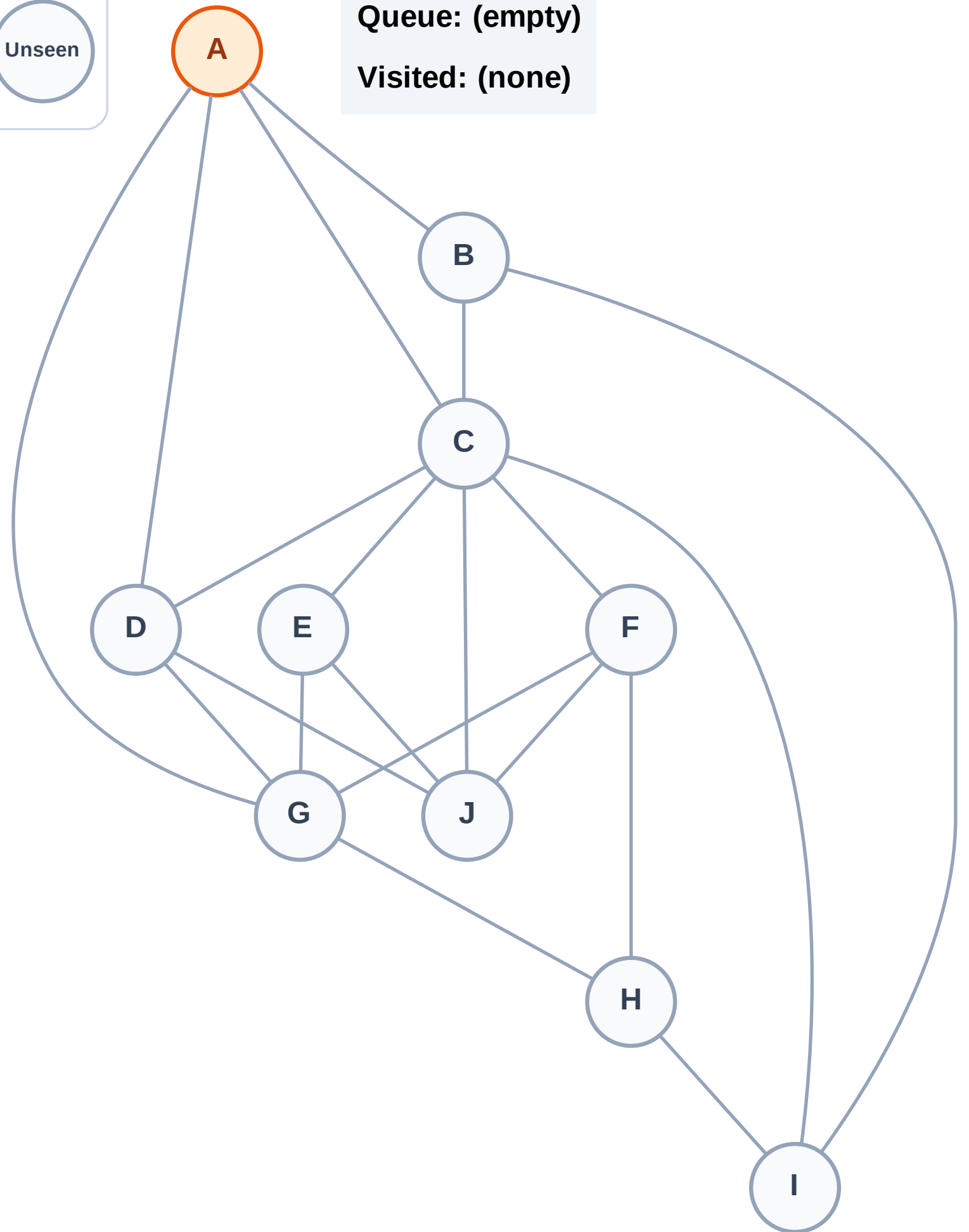
Discovered



Unseen

Queue: (empty)

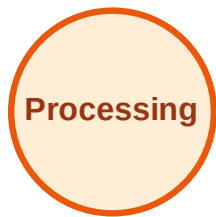
Visited: (none)



BFS Traversal

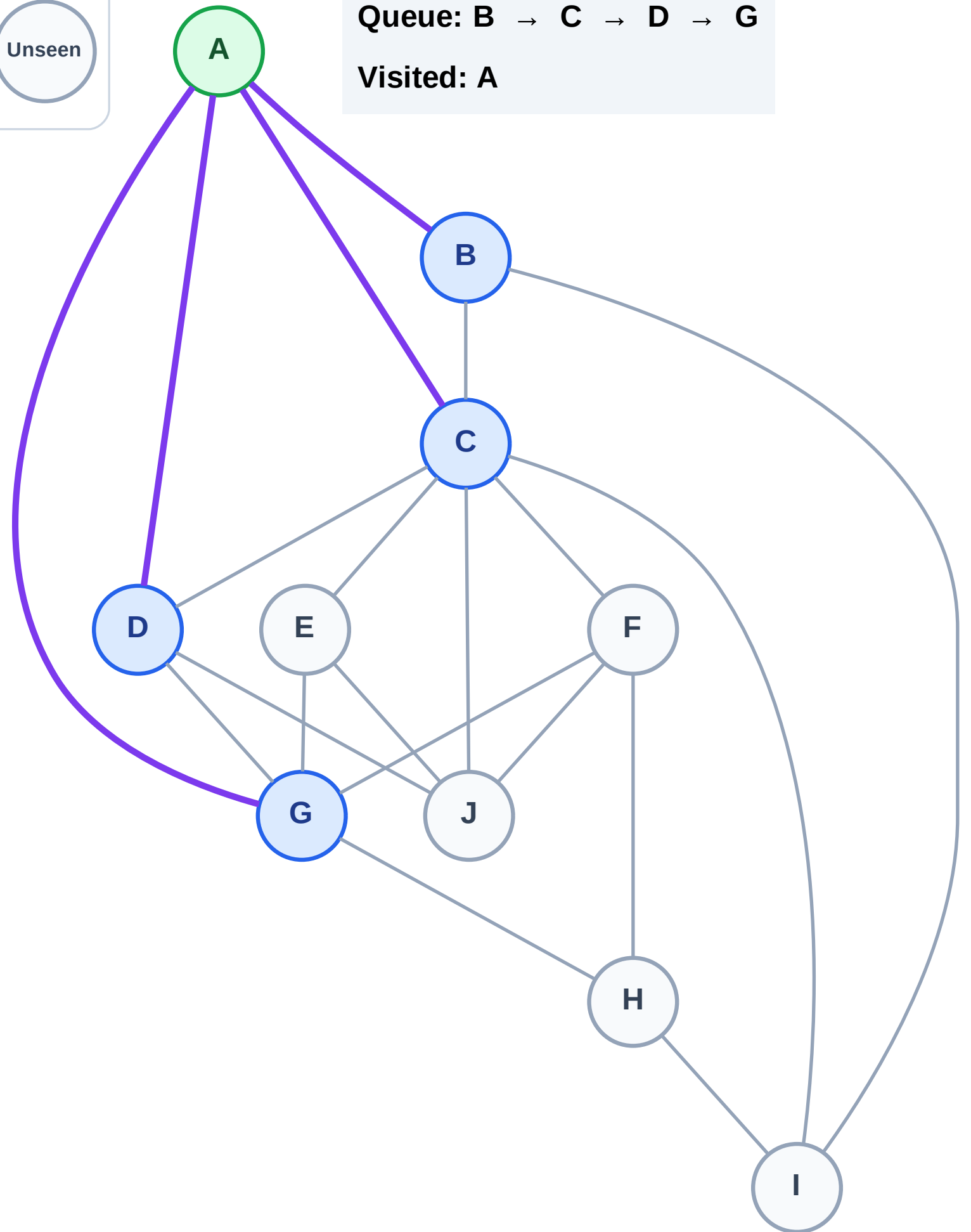
Step 3: Discover B, C, D, G from A

Legend



Queue: B → C → D → G

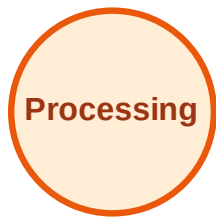
Visited: A



BFS Traversal

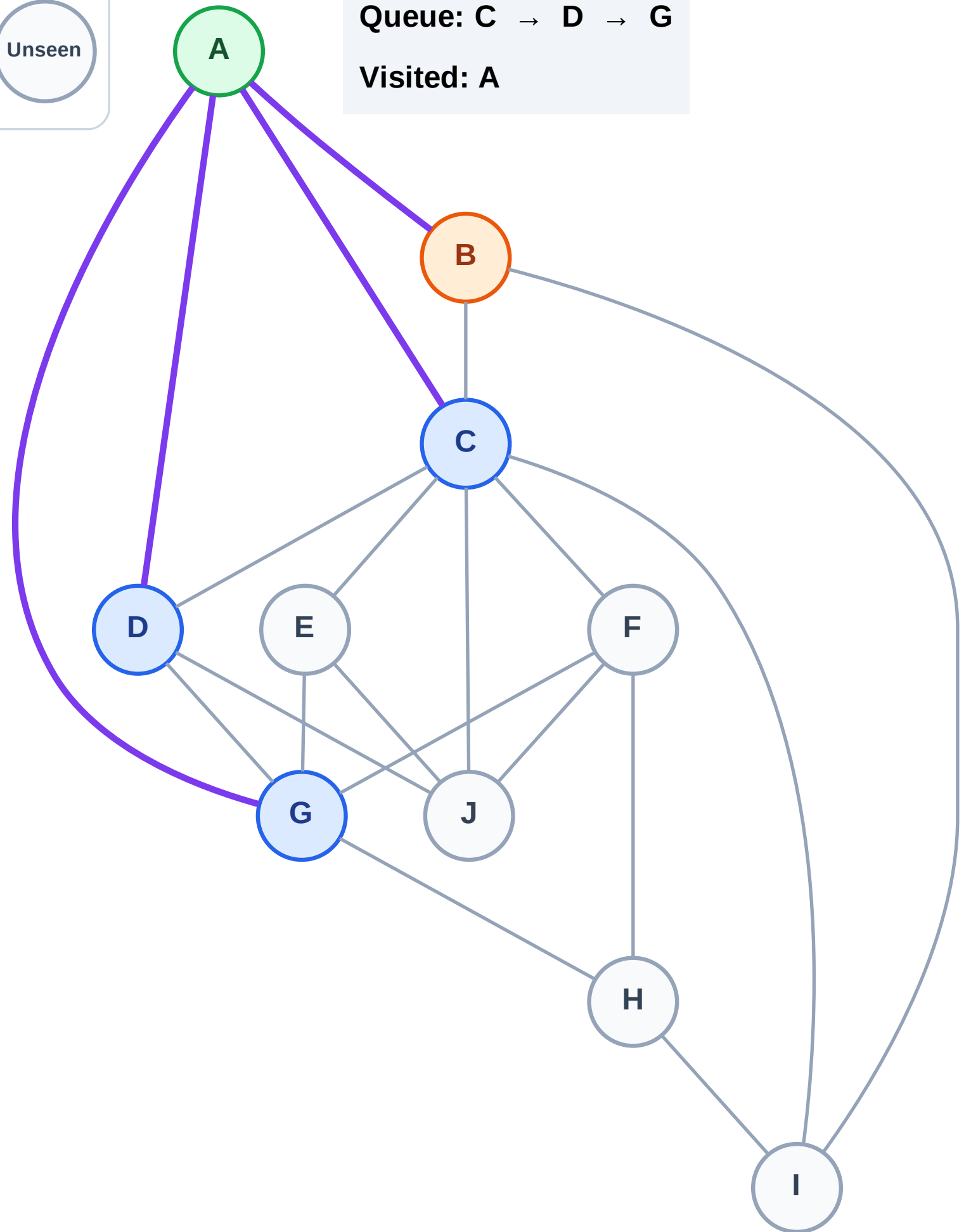
Step 4: Process node B

Legend



Queue: C → D → G

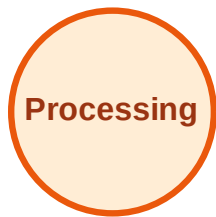
Visited: A



BFS Traversal

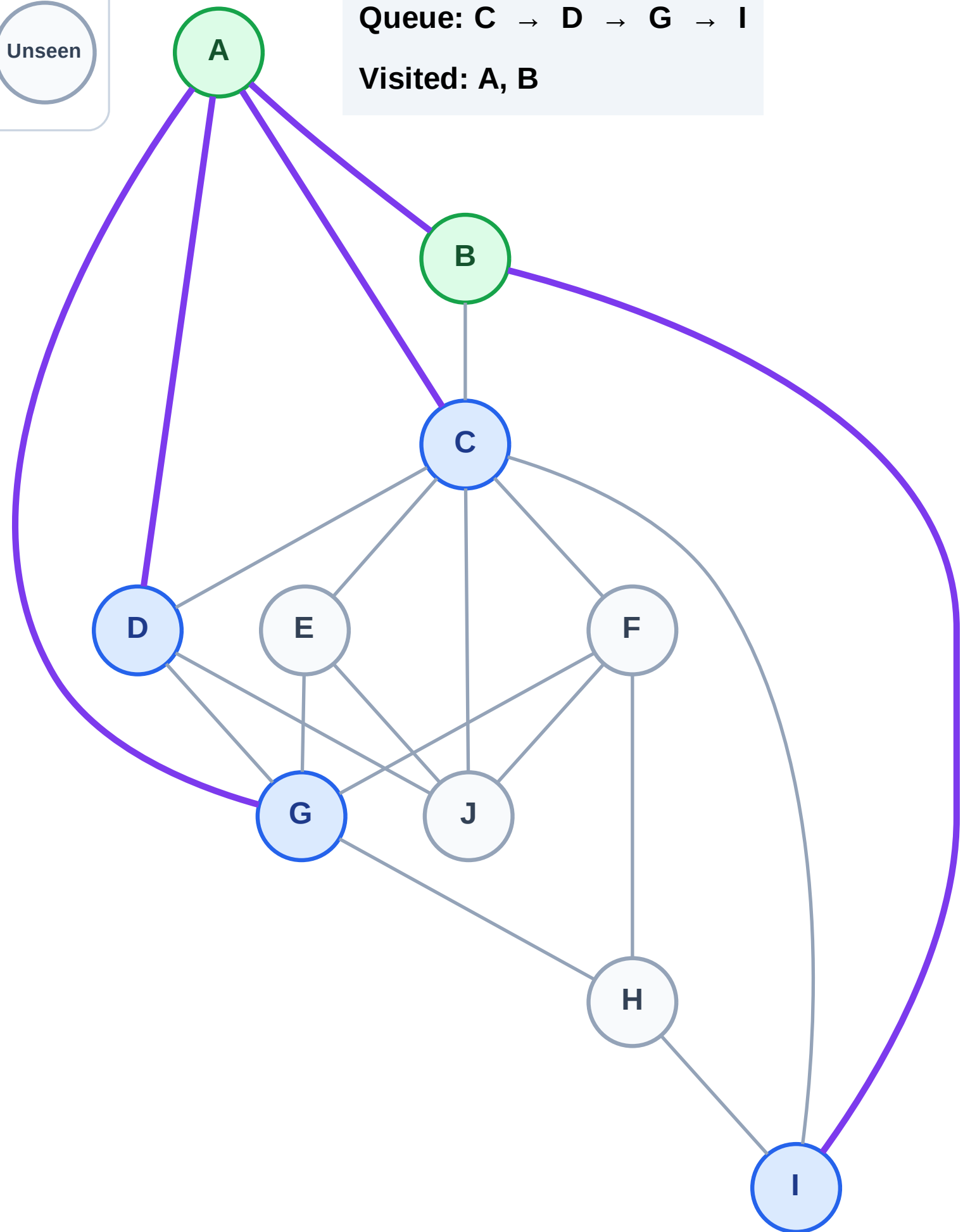
Step 5: Discover I from B

Legend



Queue: C → D → G → I

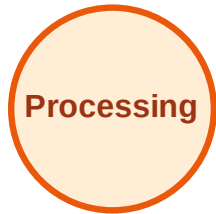
Visited: A, B



BFS Traversal

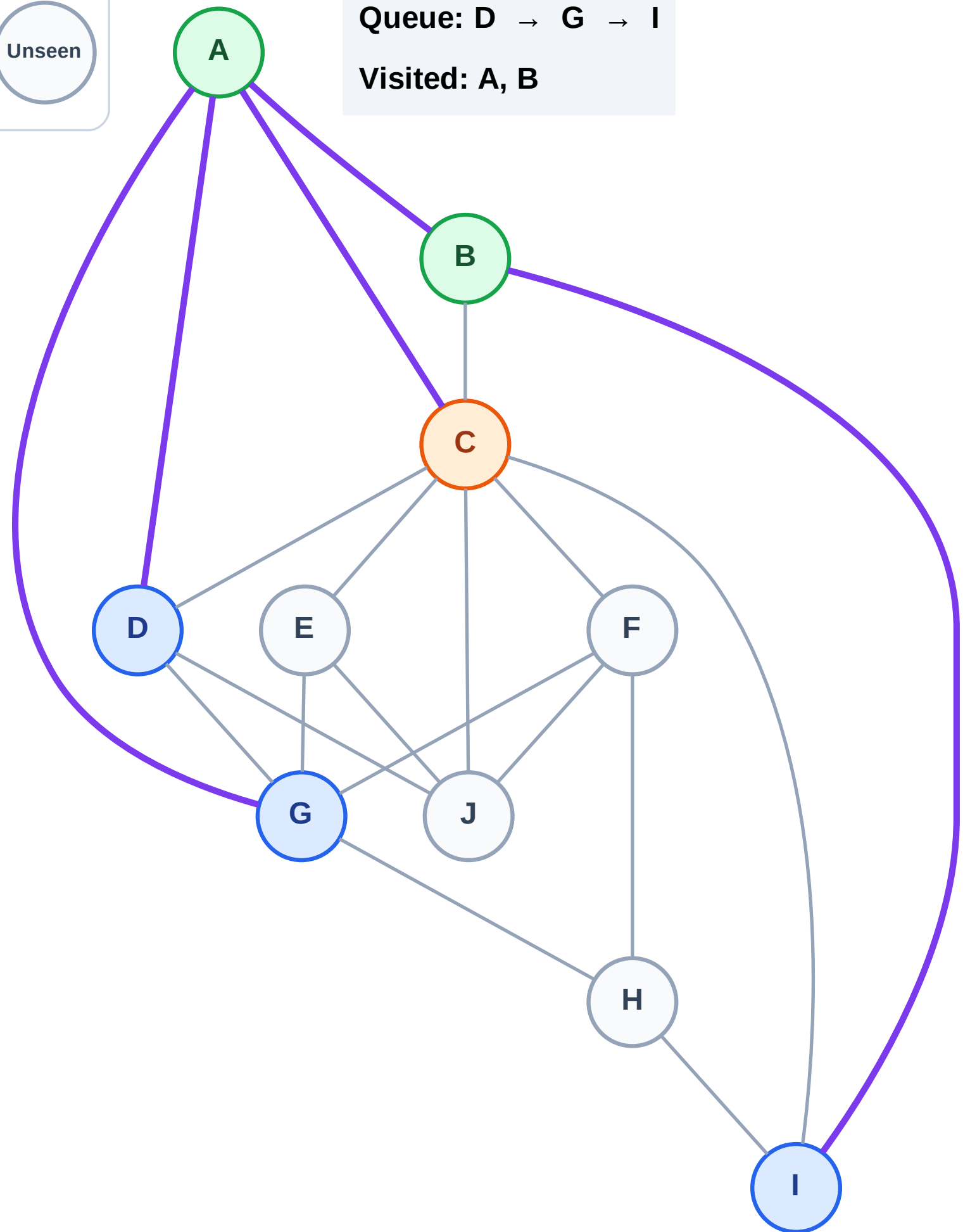
Step 6: Process node C

Legend



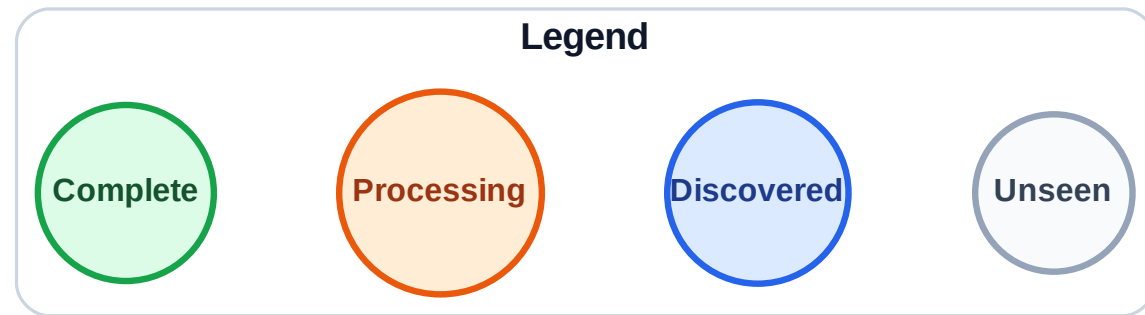
Queue: D → G → I

Visited: A, B



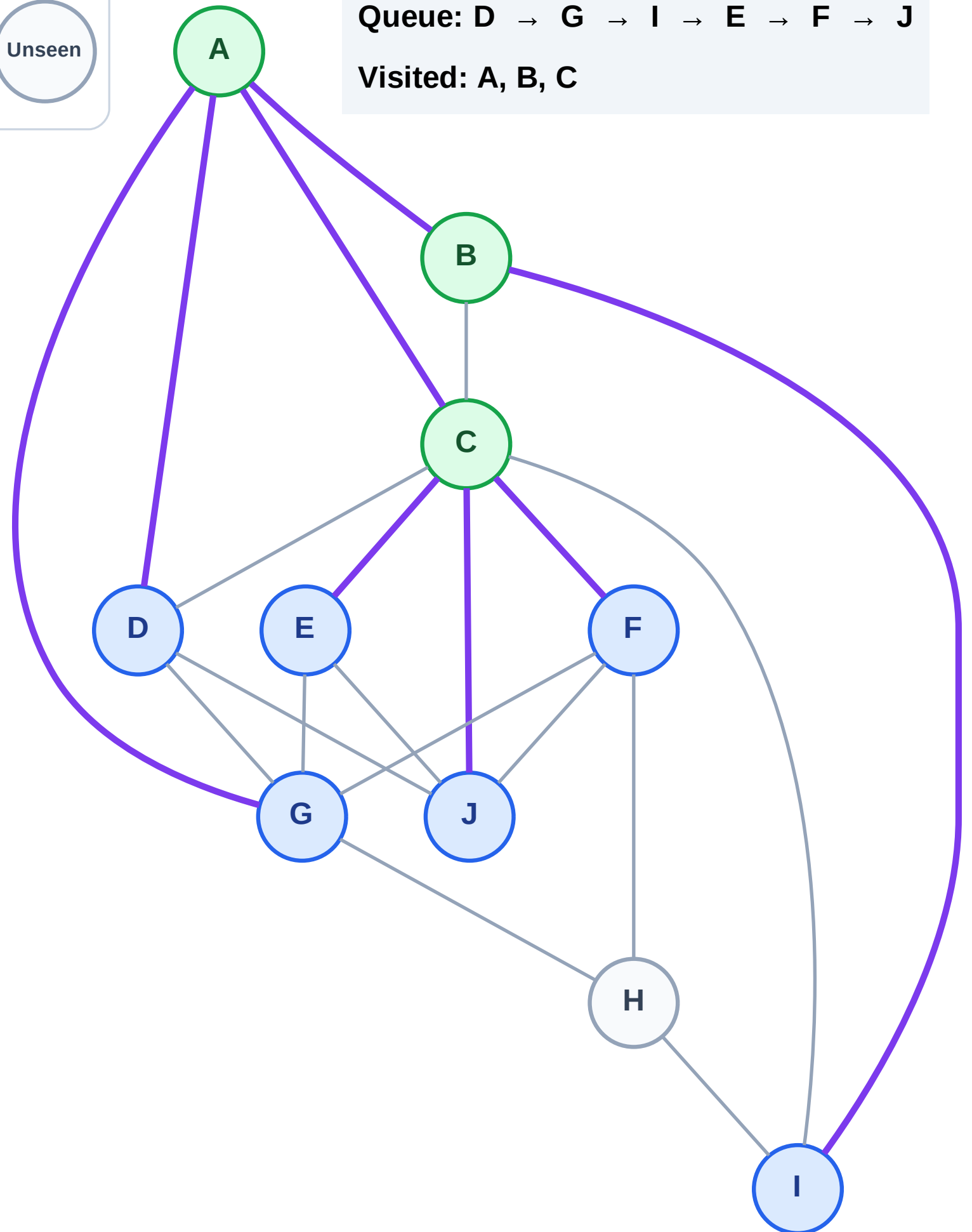
Step 7: Discover E, F, J from C

Step 7: Discover E, F, J from C



Queue: D → G → I → E → F → J

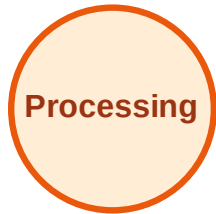
Visited: A, B, C



BFS Traversal

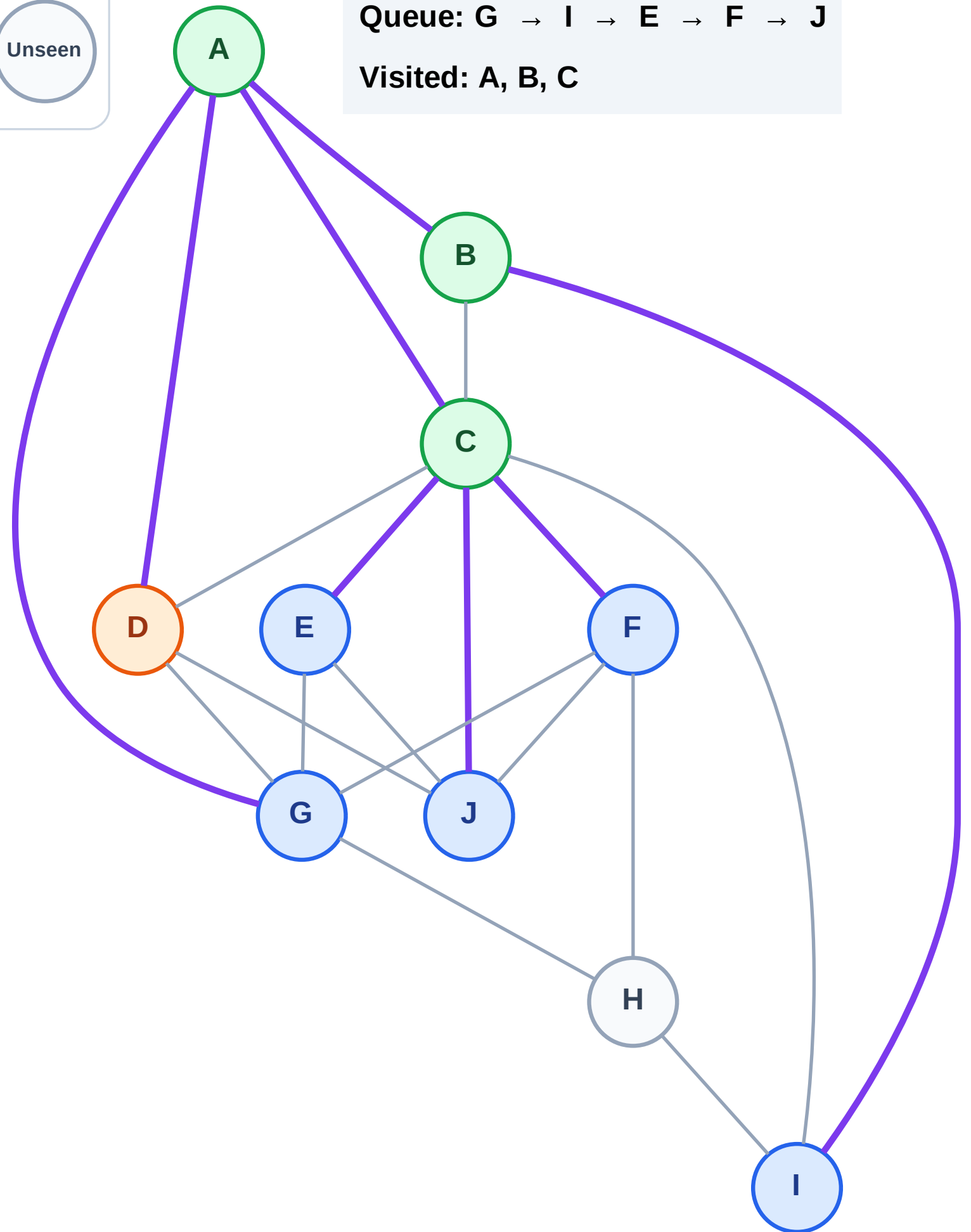
Step 8: Process node D

Legend



Queue: G → I → E → F → J

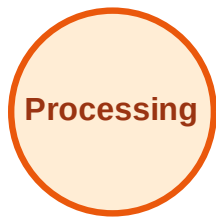
Visited: A, B, C



BFS Traversal

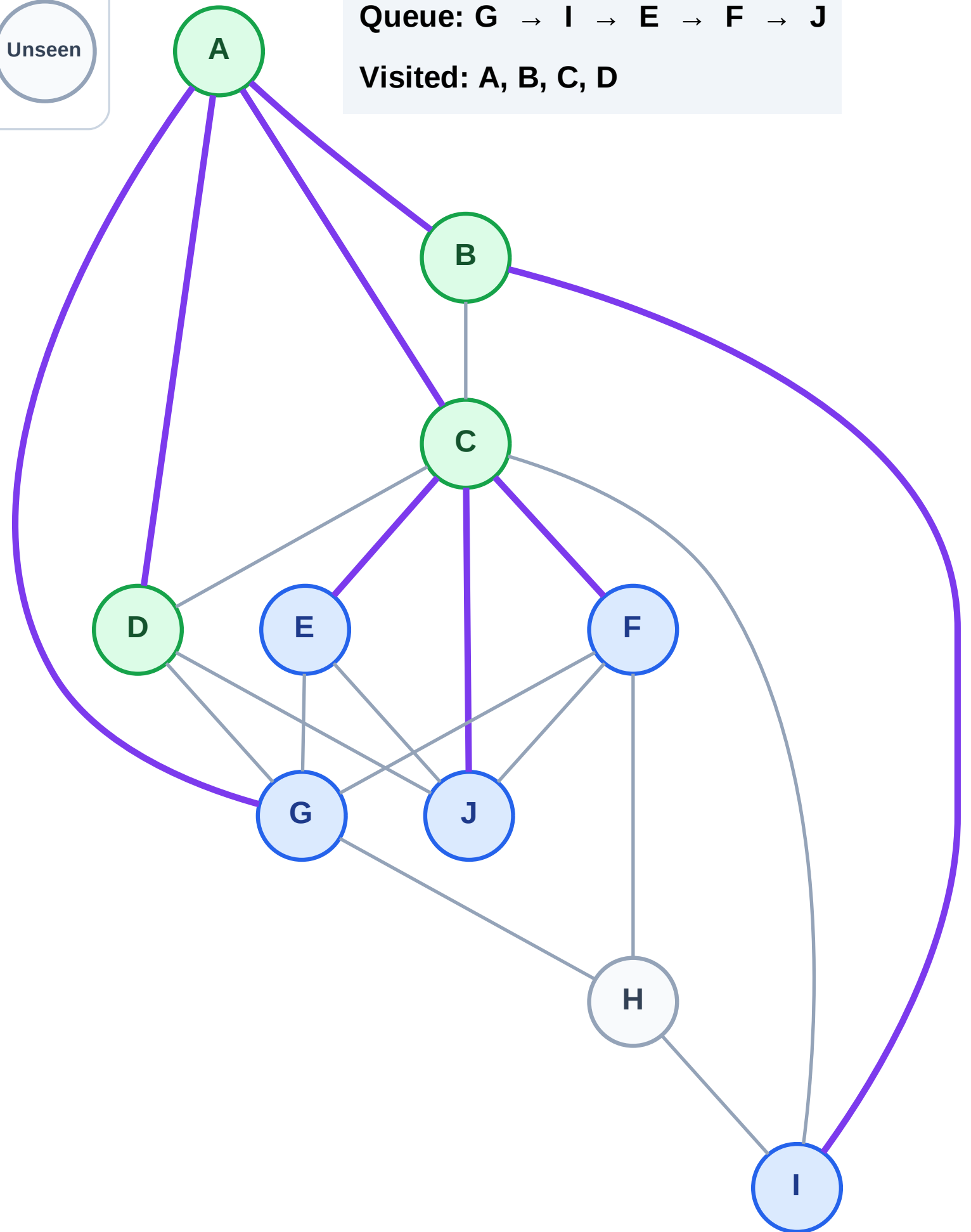
Step 9: No new neighbors from D

Legend



Queue: G → I → E → F → J

Visited: A, B, C, D



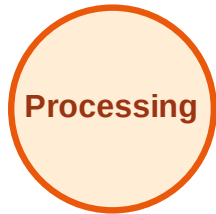
BFS Traversal

Step 10: Process node G

Legend



Complete



Processing



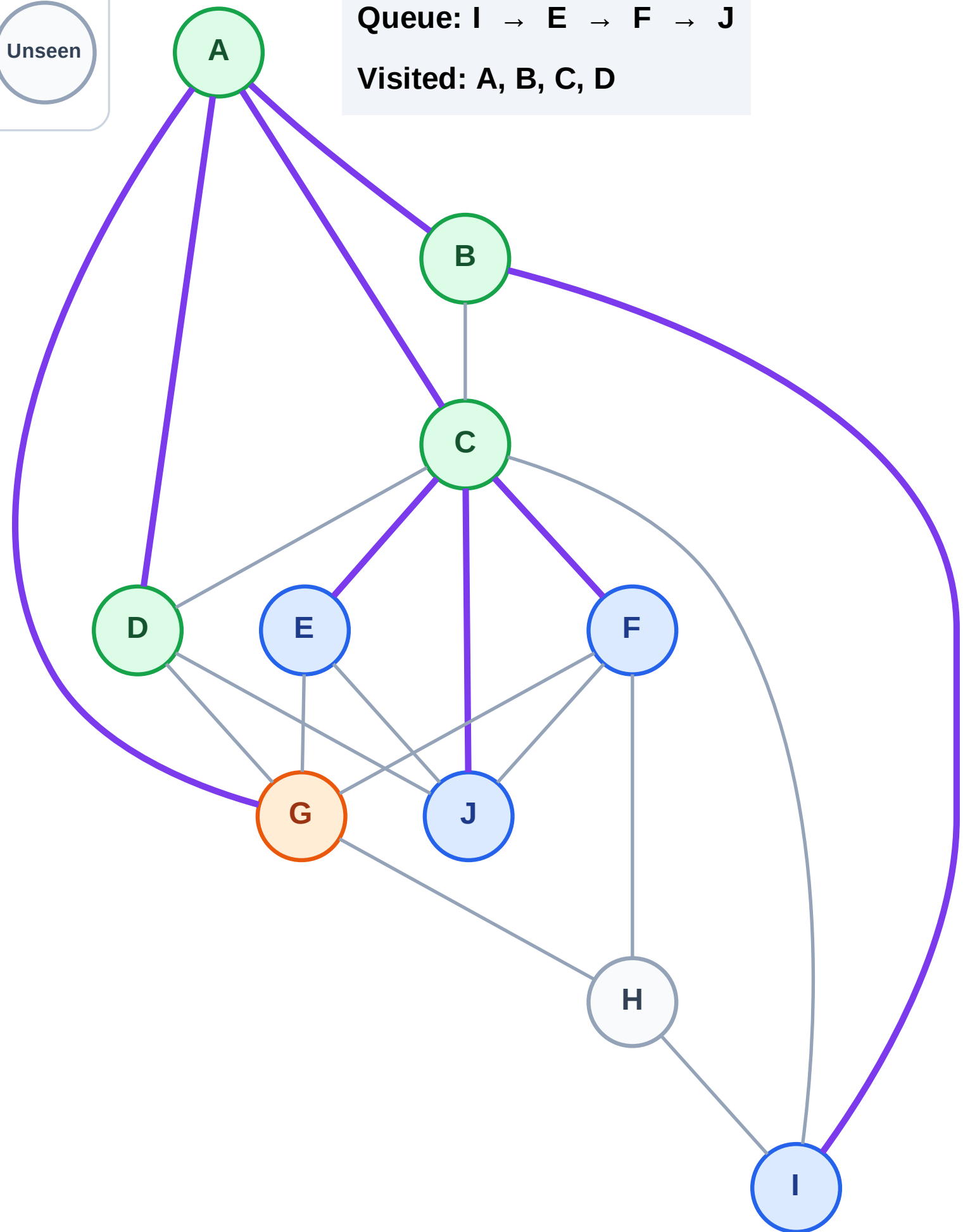
Discovered



Unseen

Queue: I → E → F → J

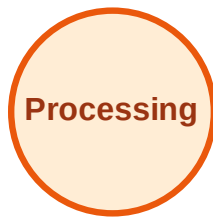
Visited: A, B, C, D



BFS Traversal

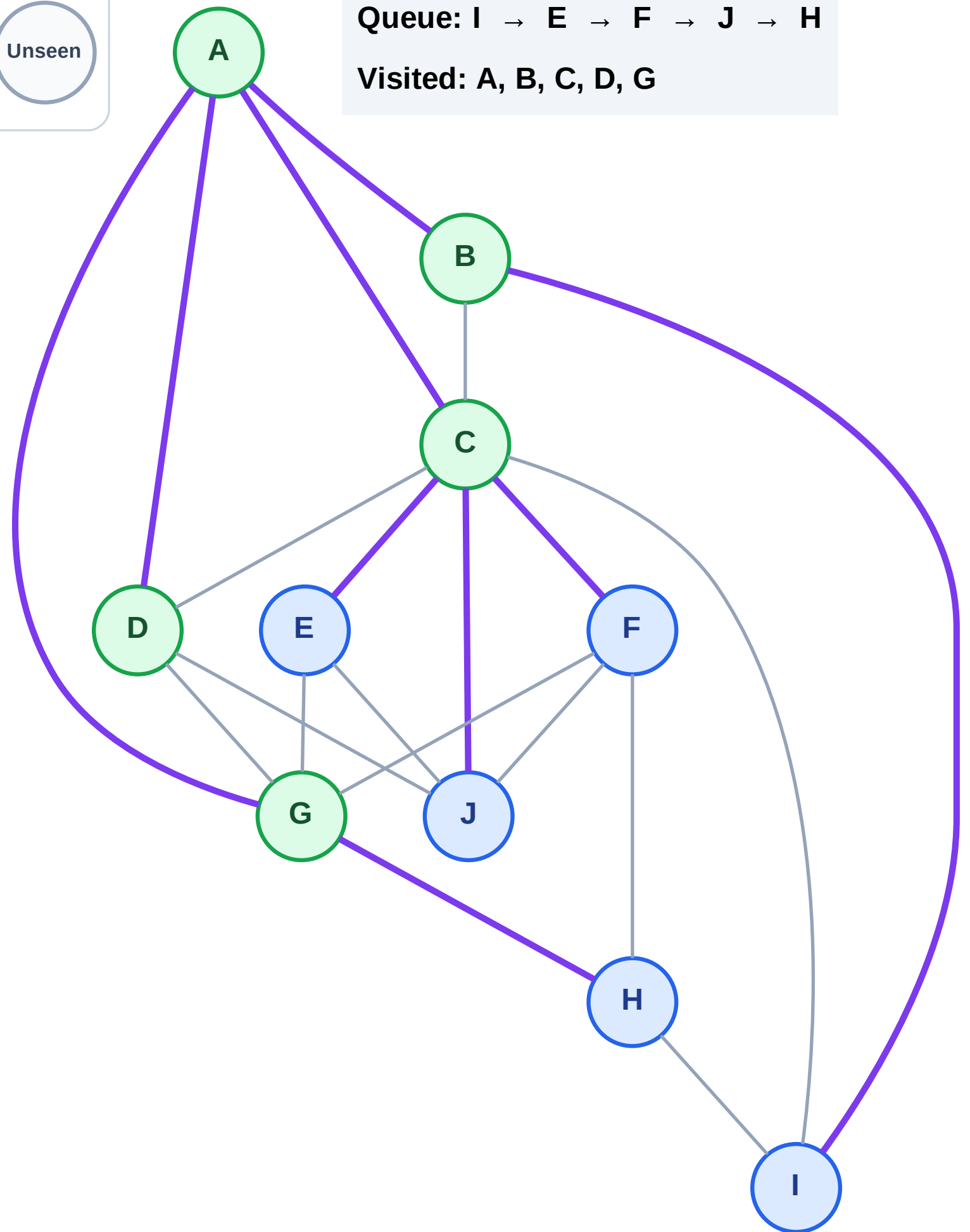
Step 11: Discover H from G

Legend



Queue: I → E → F → J → H

Visited: A, B, C, D, G



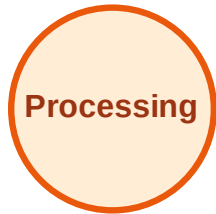
BFS Traversal

Step 12: Process node I

Legend



Complete



Processing



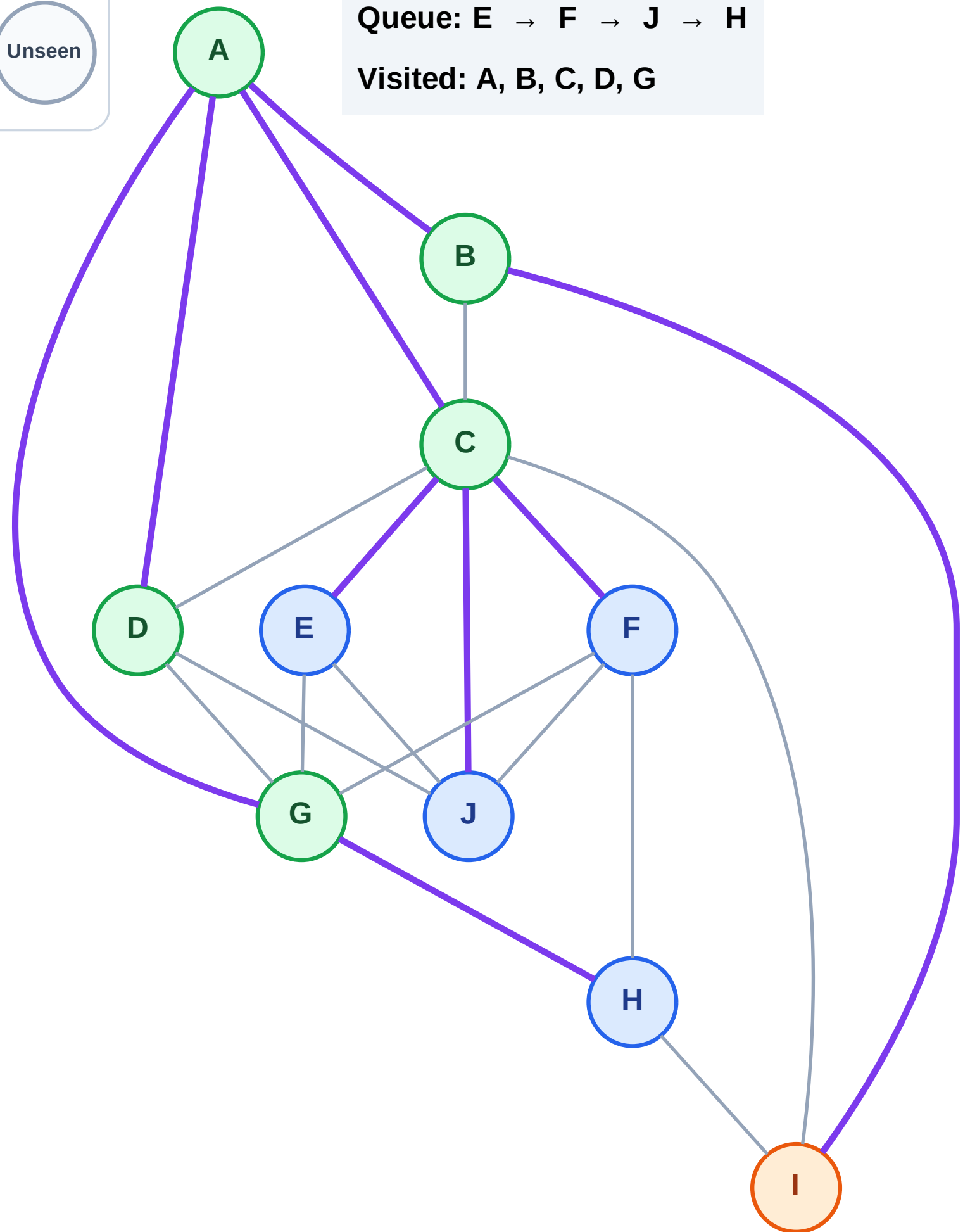
Discovered



Unseen

Queue: E → F → J → H

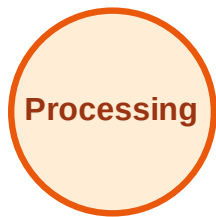
Visited: A, B, C, D, G



BFS Traversal

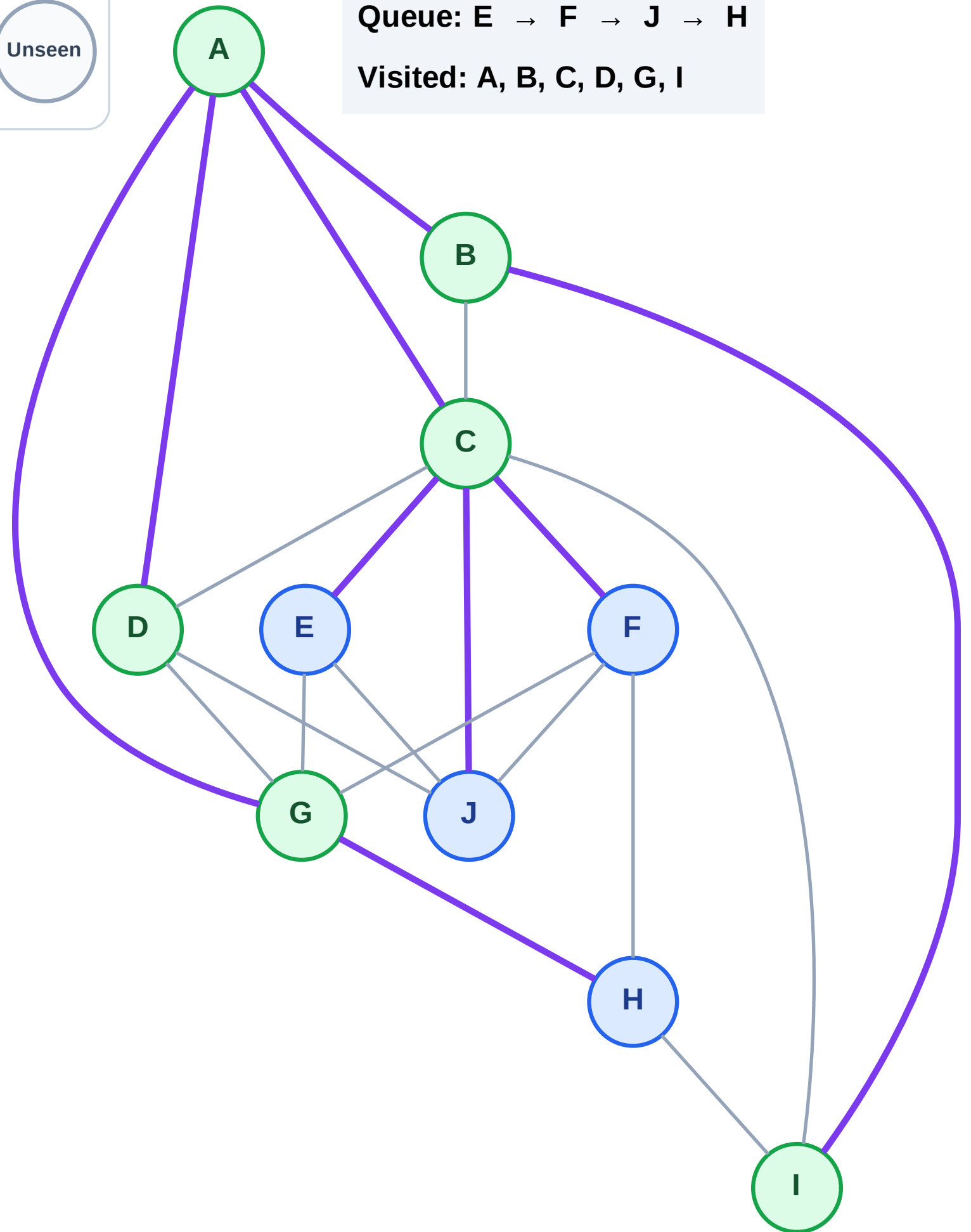
Step 13: No new neighbors from I

Legend



Queue: E → F → J → H

Visited: A, B, C, D, G, I



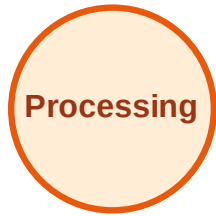
BFS Traversal

Step 14: Process node E

Legend



Complete



Processing



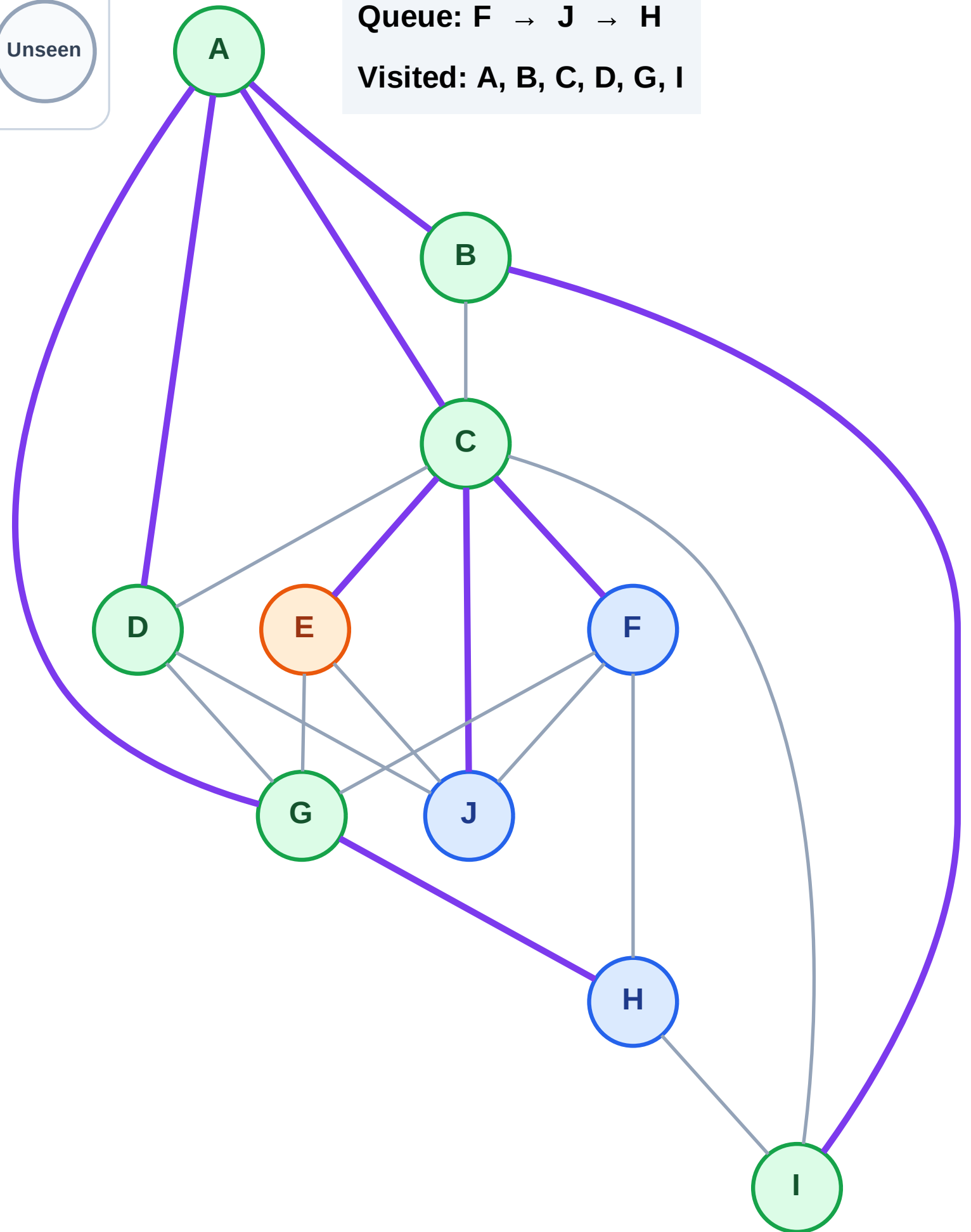
Discovered



Unseen

Queue: F → J → H

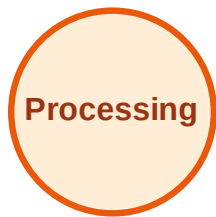
Visited: A, B, C, D, G, I



BFS Traversal

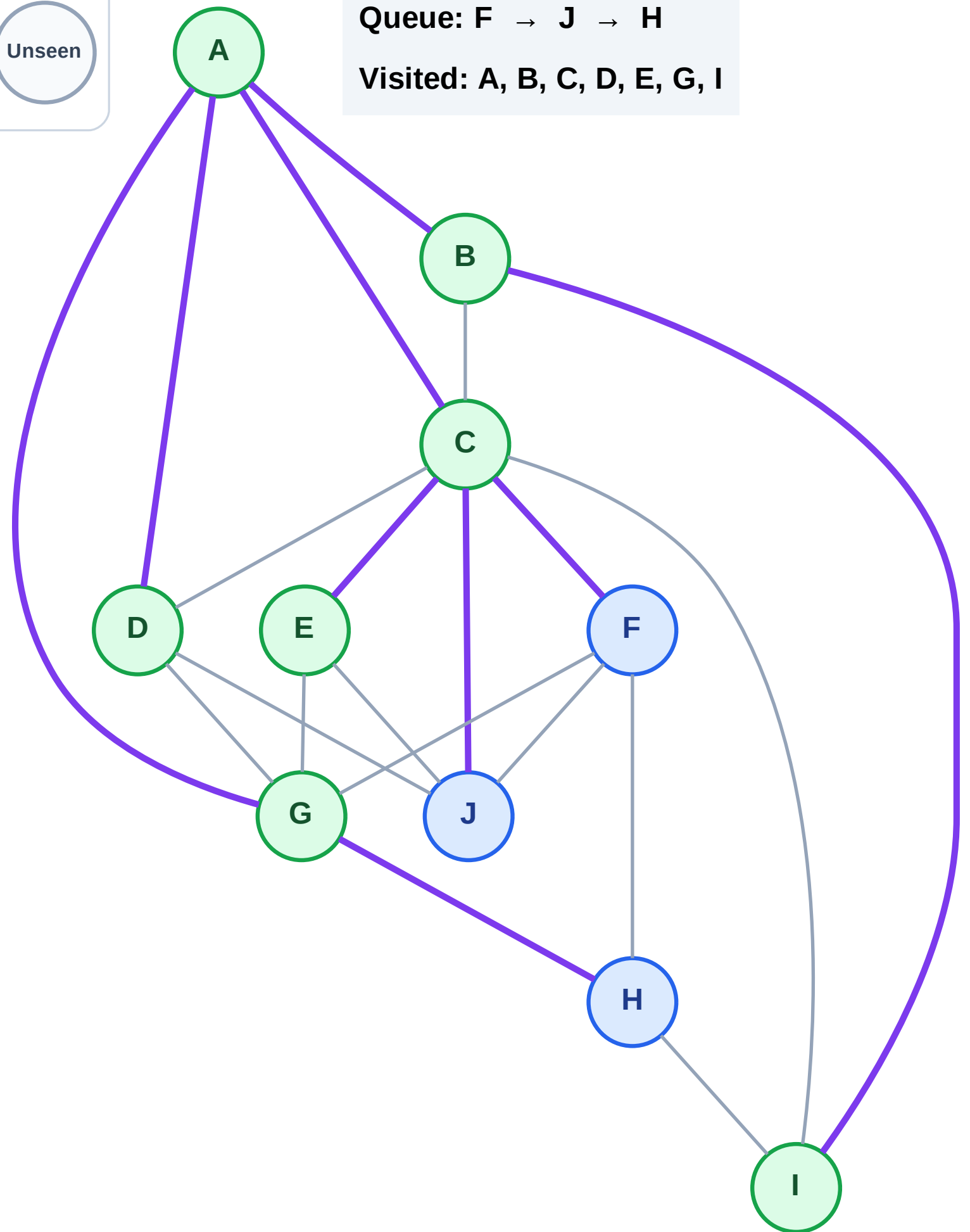
Step 15: No new neighbors from E

Legend



Queue: F → J → H

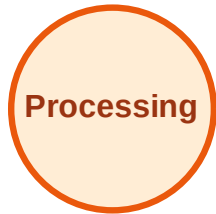
Visited: A, B, C, D, E, G, I



BFS Traversal

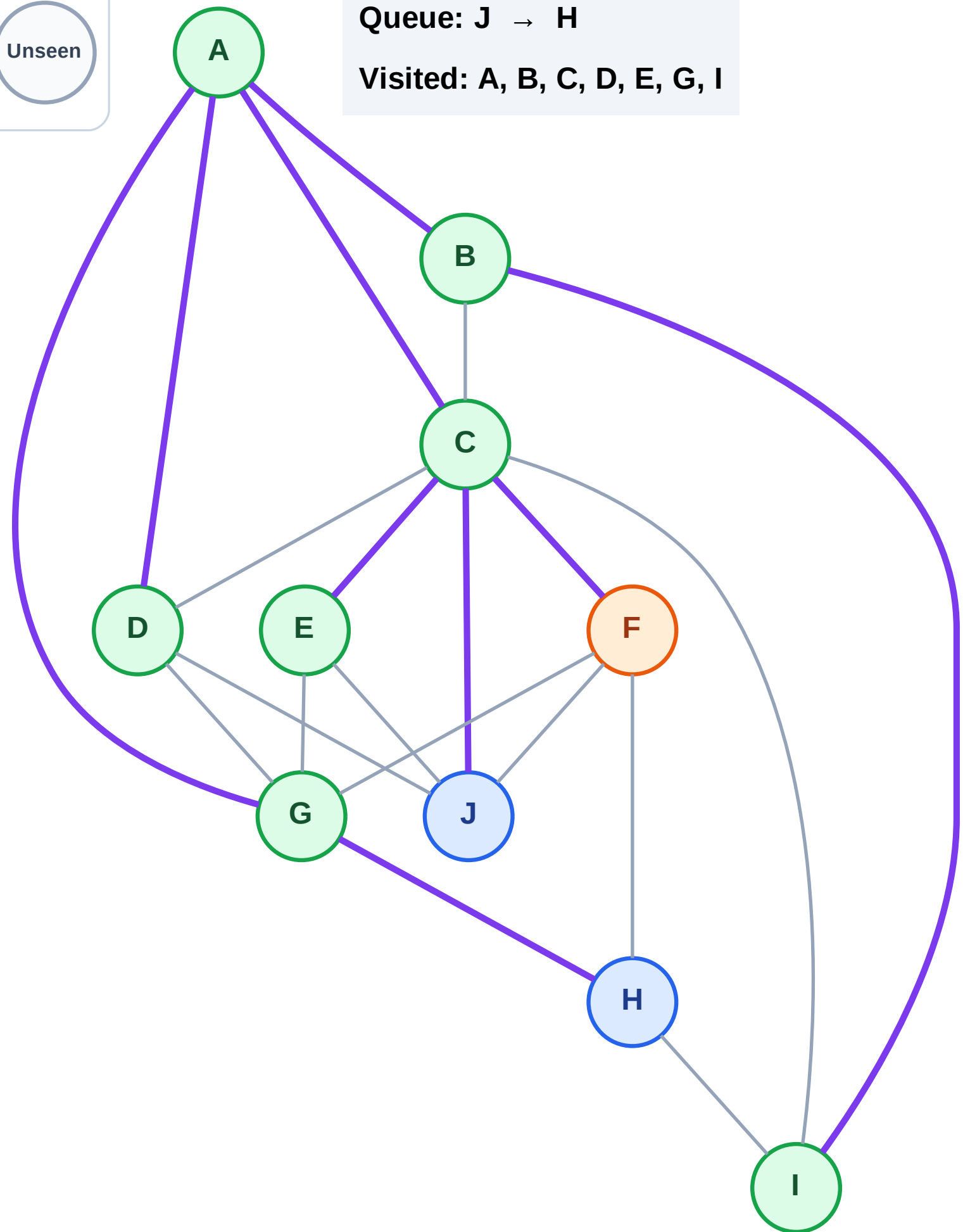
Step 16: Process node F

Legend



Queue: J → H

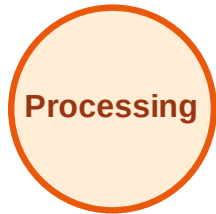
Visited: A, B, C, D, E, G, I



BFS Traversal

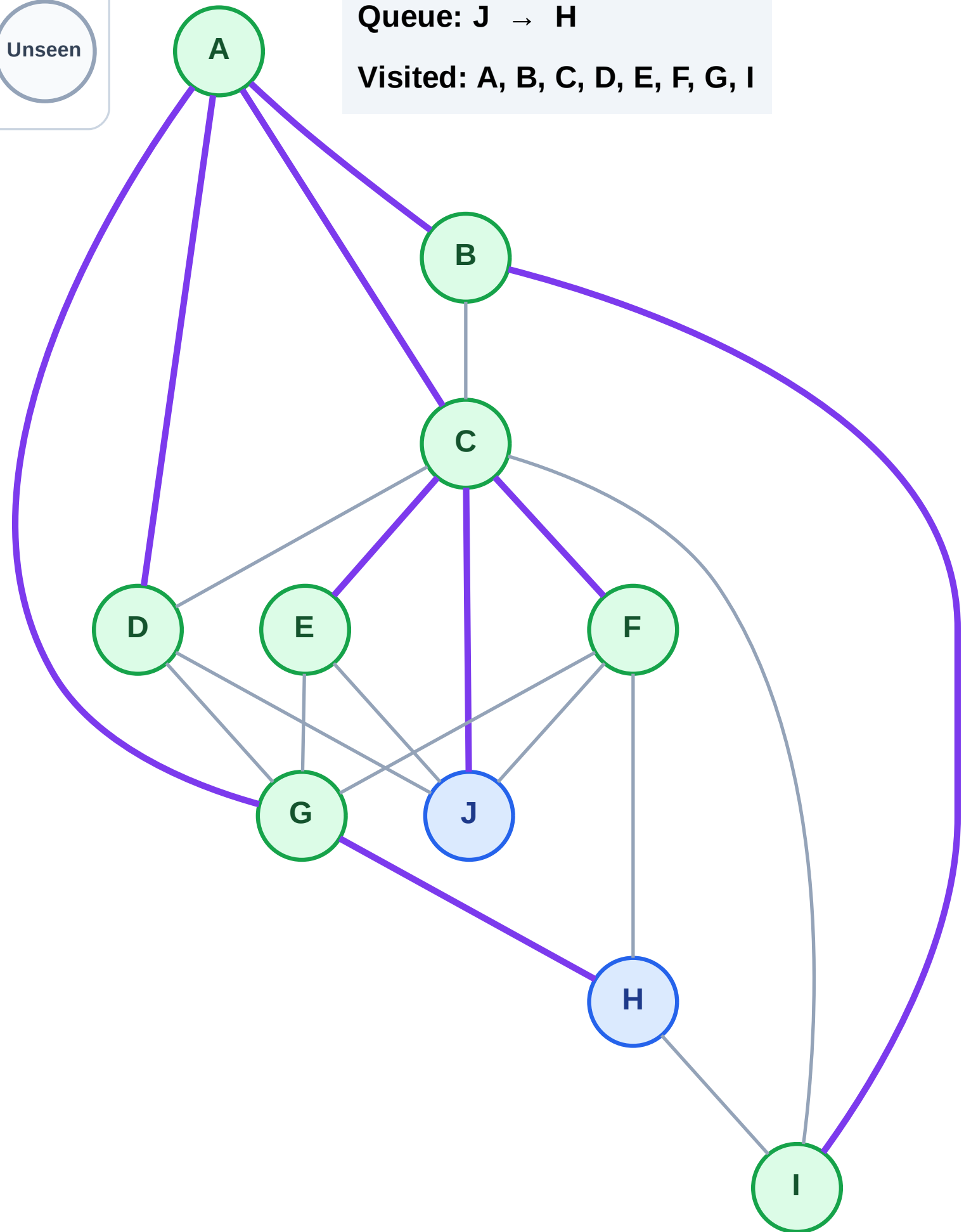
Step 17: No new neighbors from F

Legend



Queue: J → H

Visited: A, B, C, D, E, F, G, I



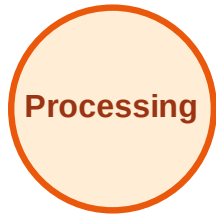
BFS Traversal

Step 18: Process node J

Legend



Complete



Processing



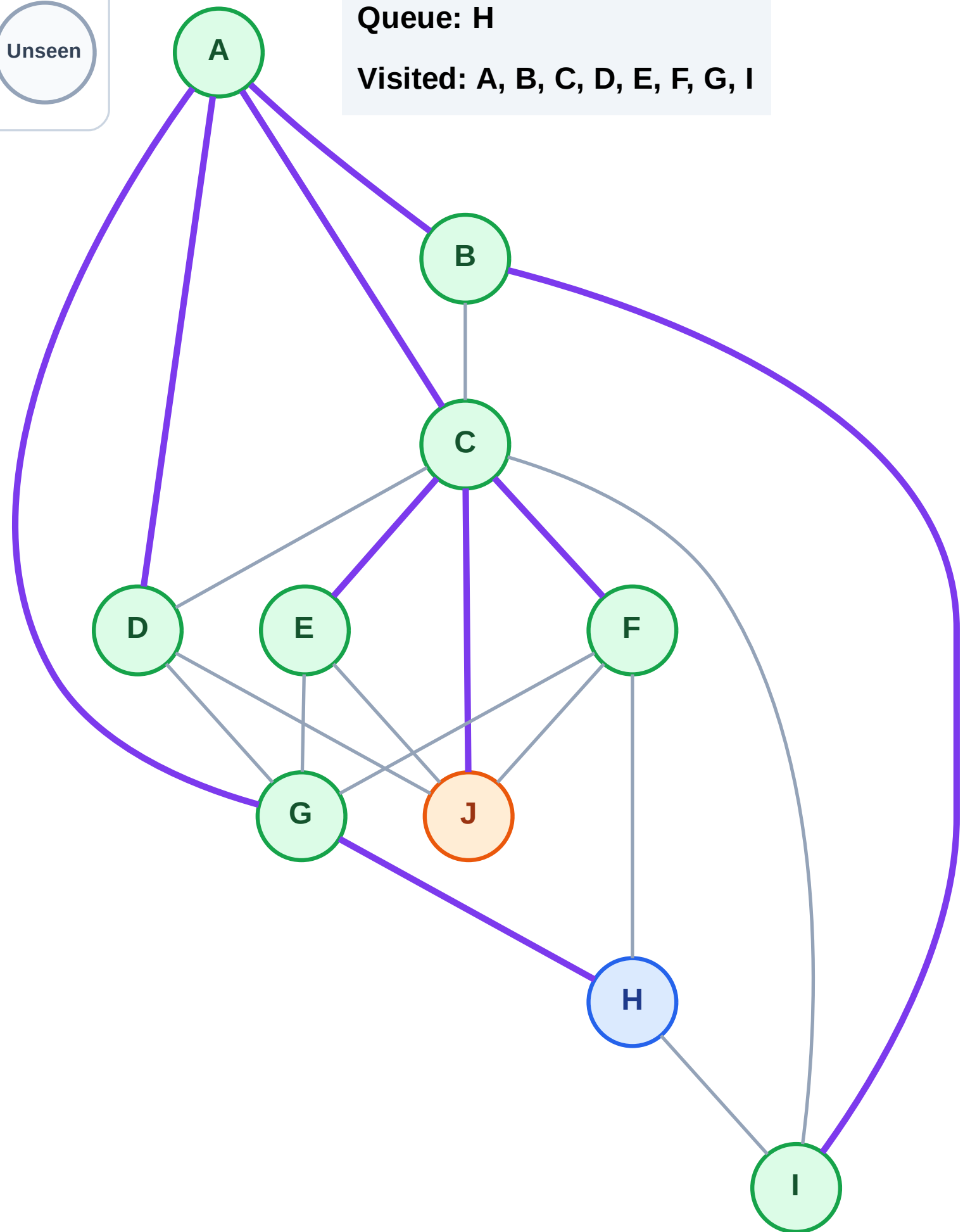
Discovered



Unseen

Queue: H

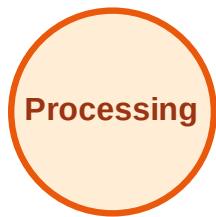
Visited: A, B, C, D, E, F, G, I



BFS Traversal

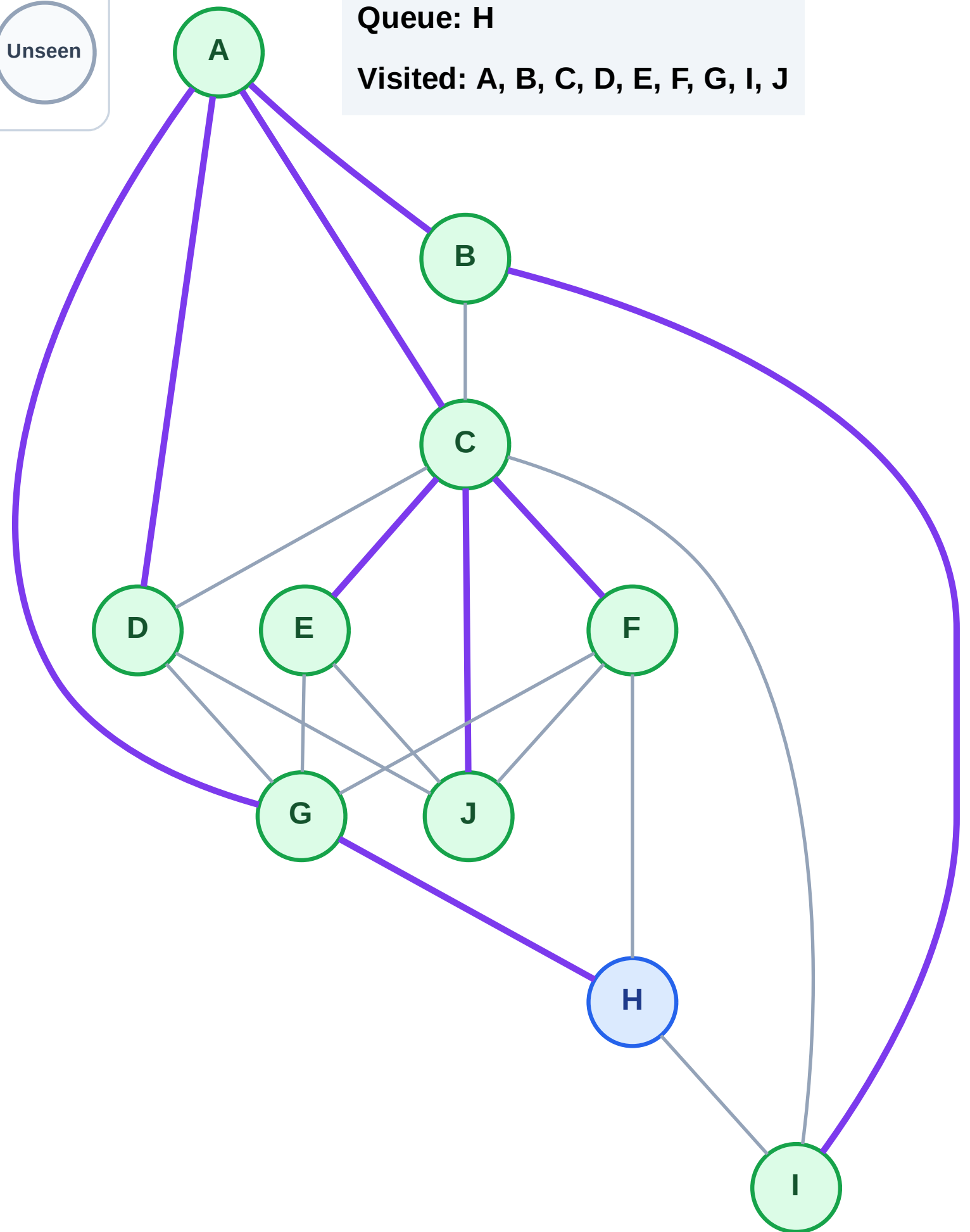
Step 19: No new neighbors from J

Legend



Queue: H

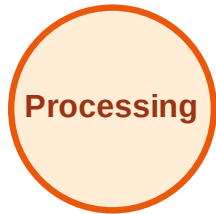
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

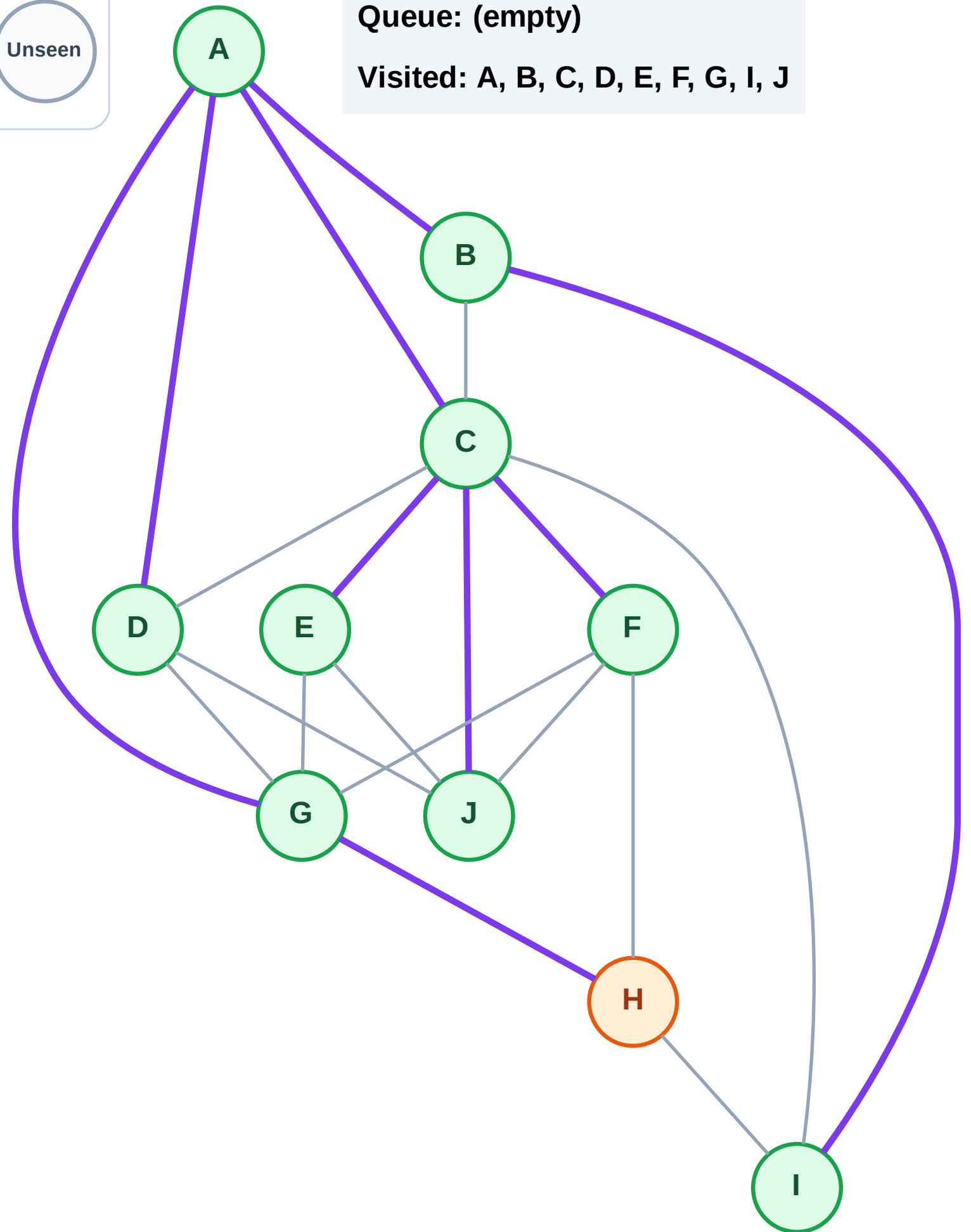
Step 20: Process node H

Legend



Queue: (empty)

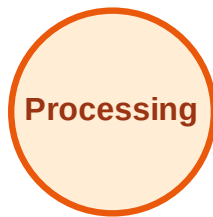
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

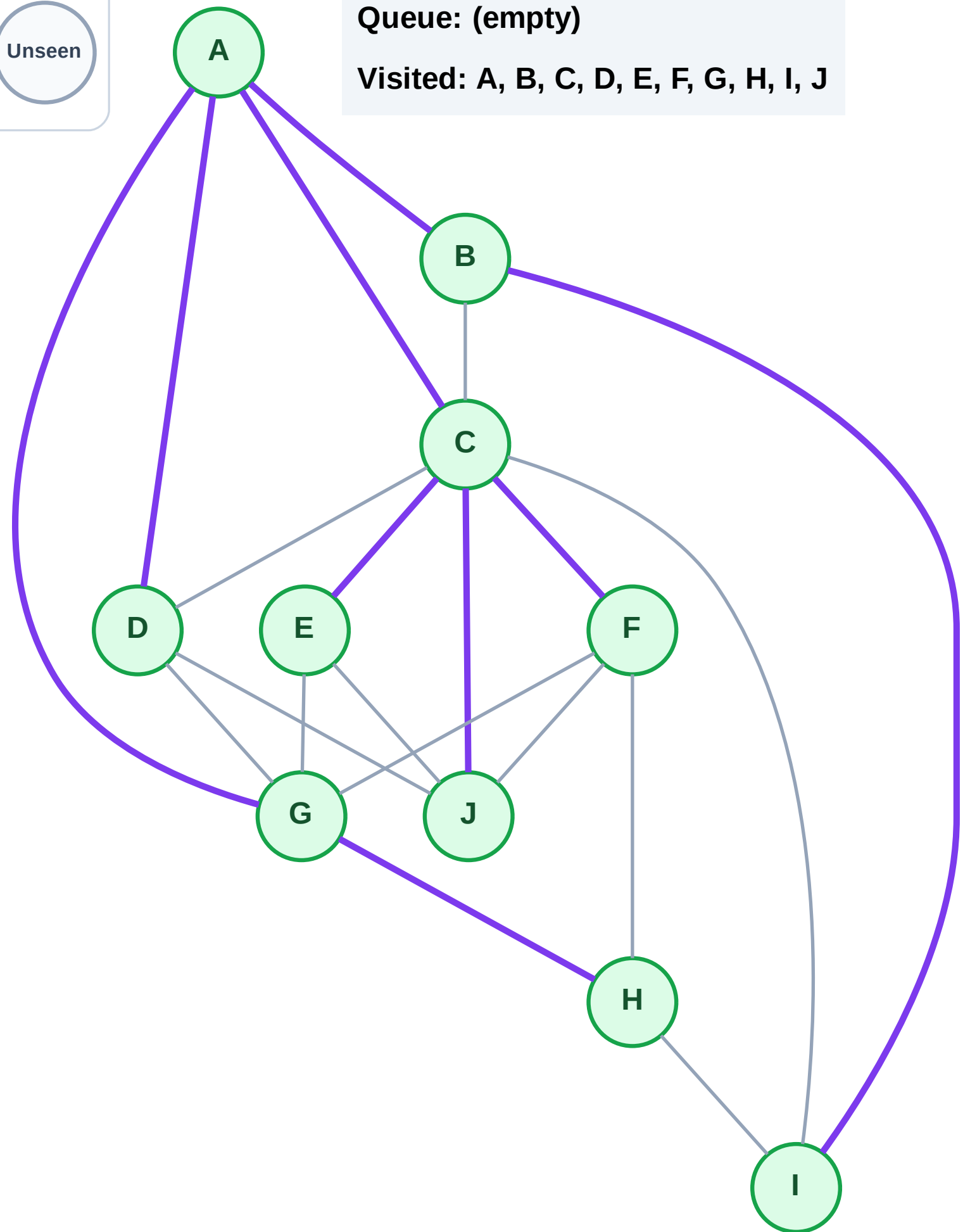
Step 21: No new neighbors from H

Legend



Queue: (empty)

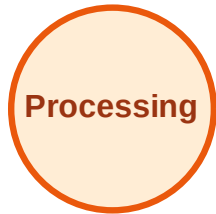
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

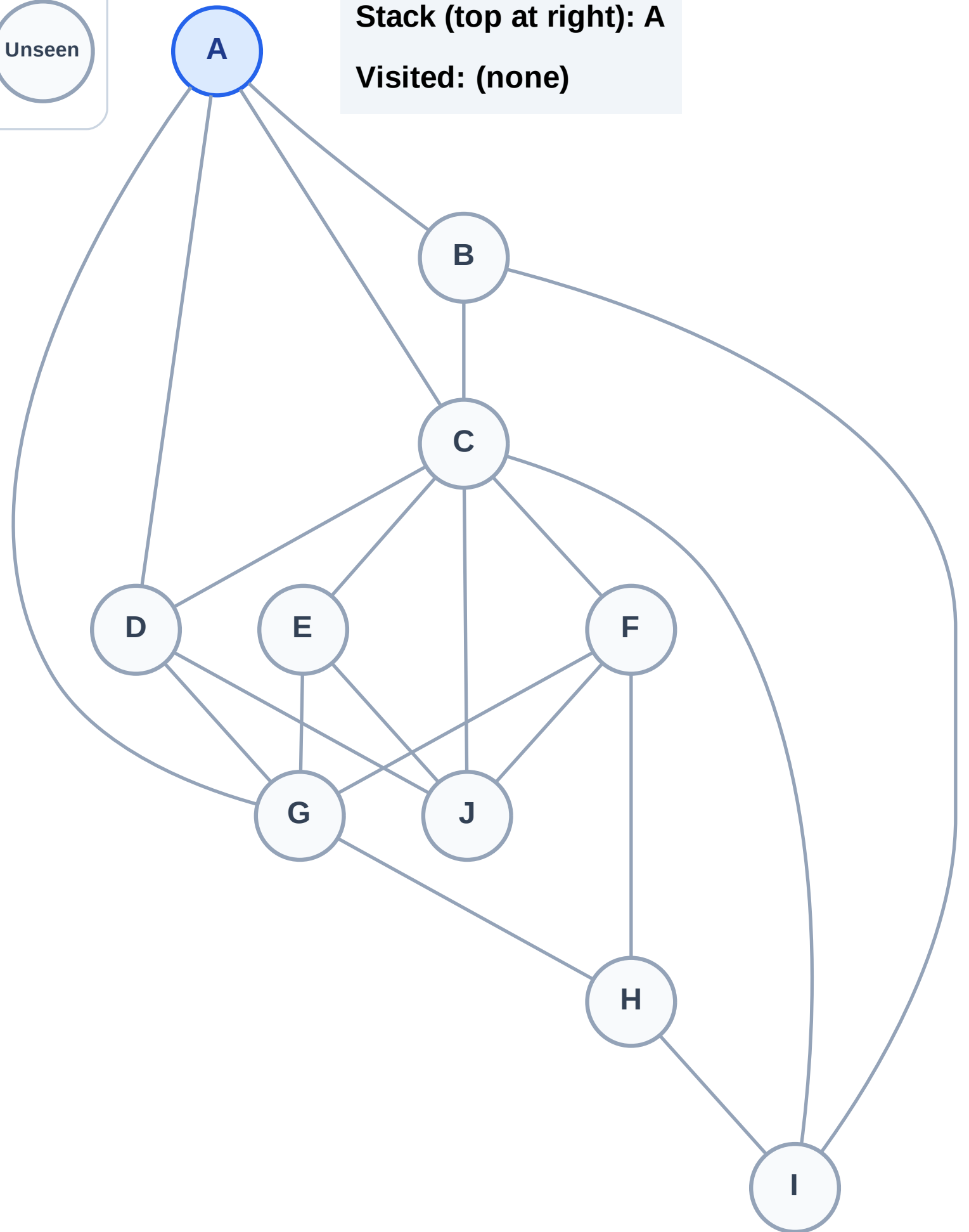
Step 1: Start at A

Legend



Stack (top at right): A

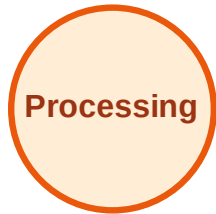
Visited: (none)



DFS Traversal

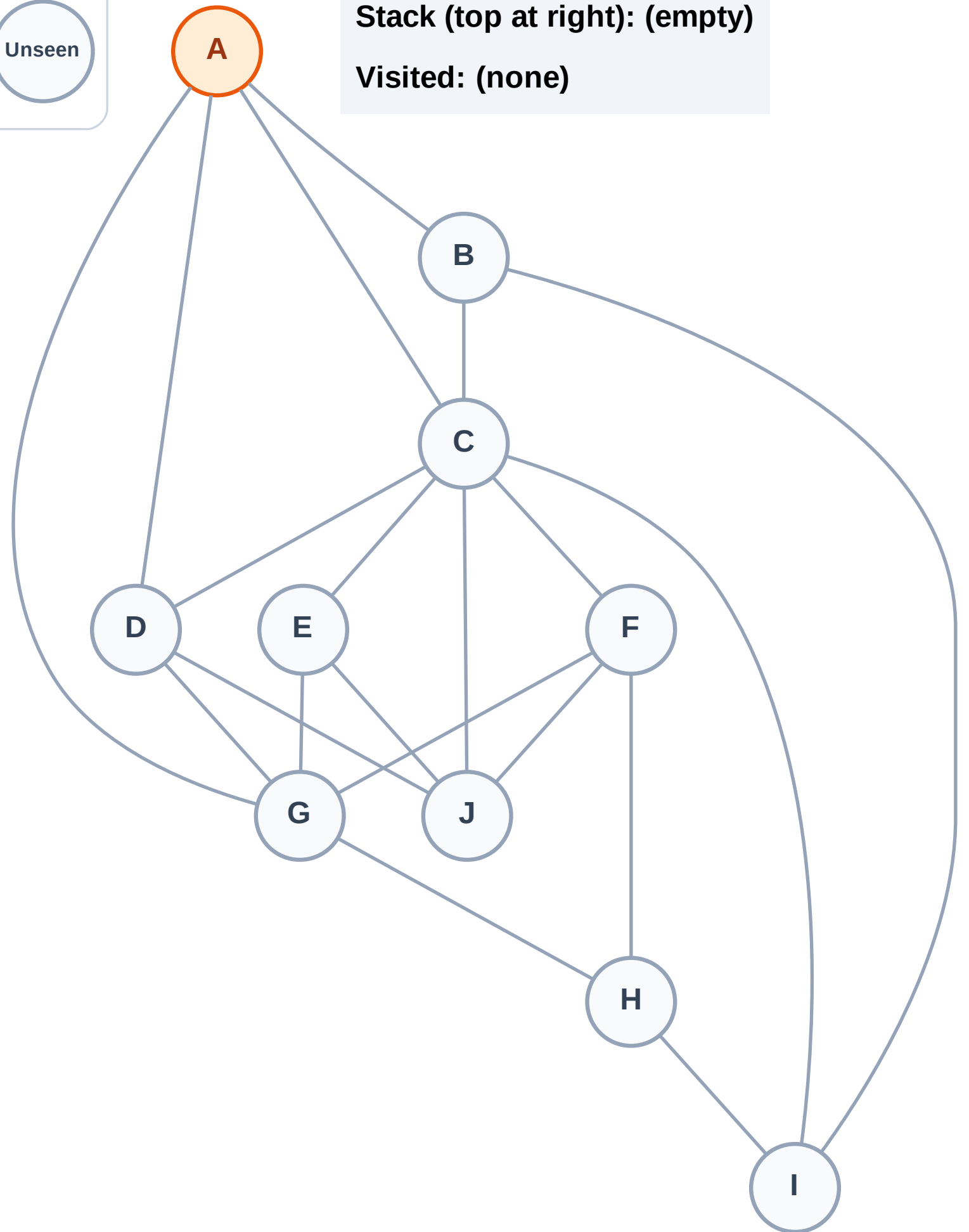
Step 2: Process node A

Legend



Stack (top at right): (empty)

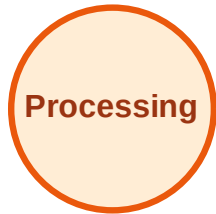
Visited: (none)



DFS Traversal

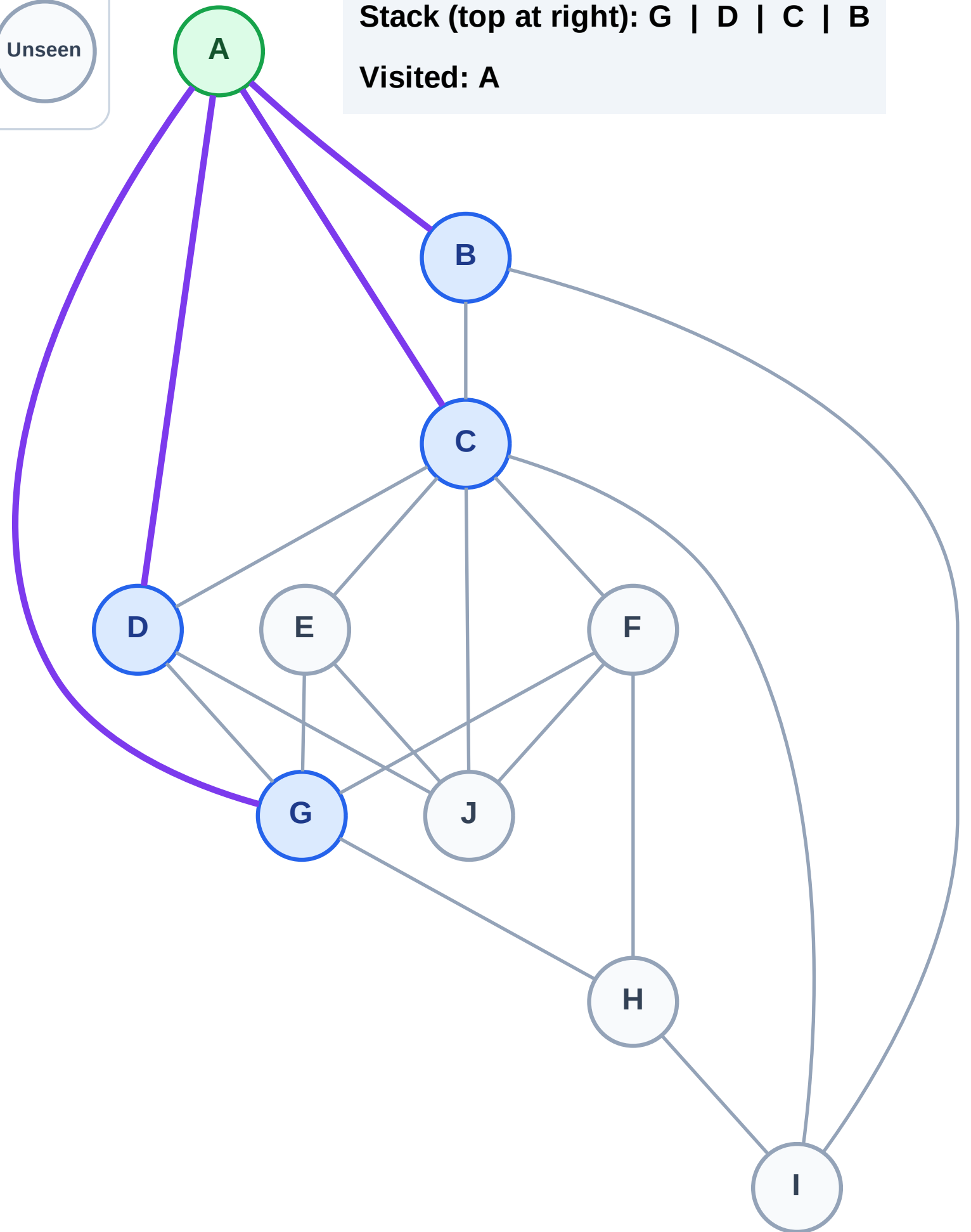
Step 3: Discover G, D, C, B from A

Legend



Stack (top at right): G | D | C | B

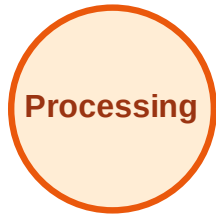
Visited: A



DFS Traversal

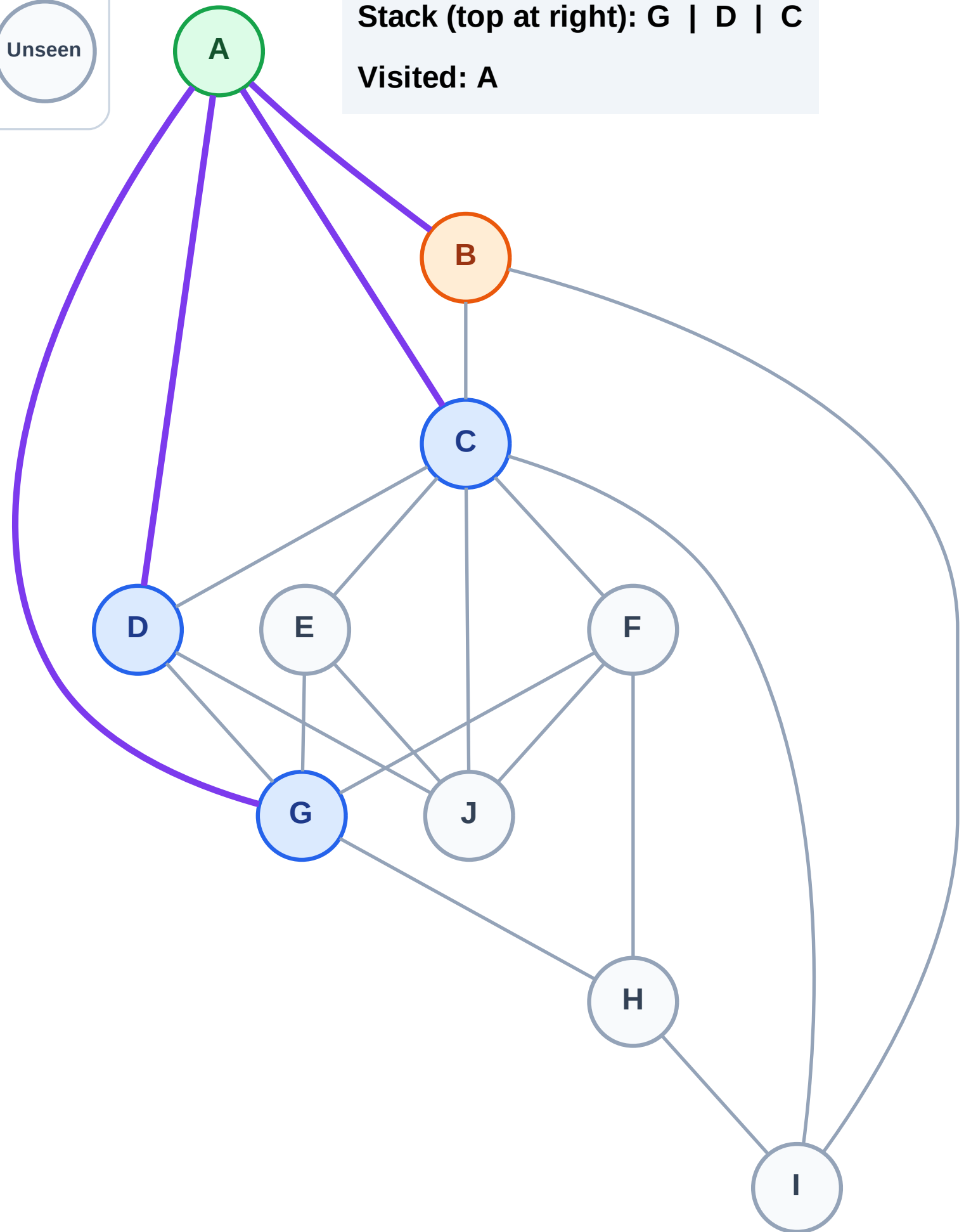
Step 4: Process node B

Legend



Stack (top at right): G | D | C

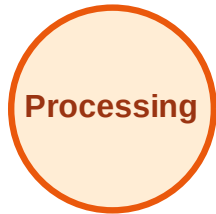
Visited: A



DFS Traversal

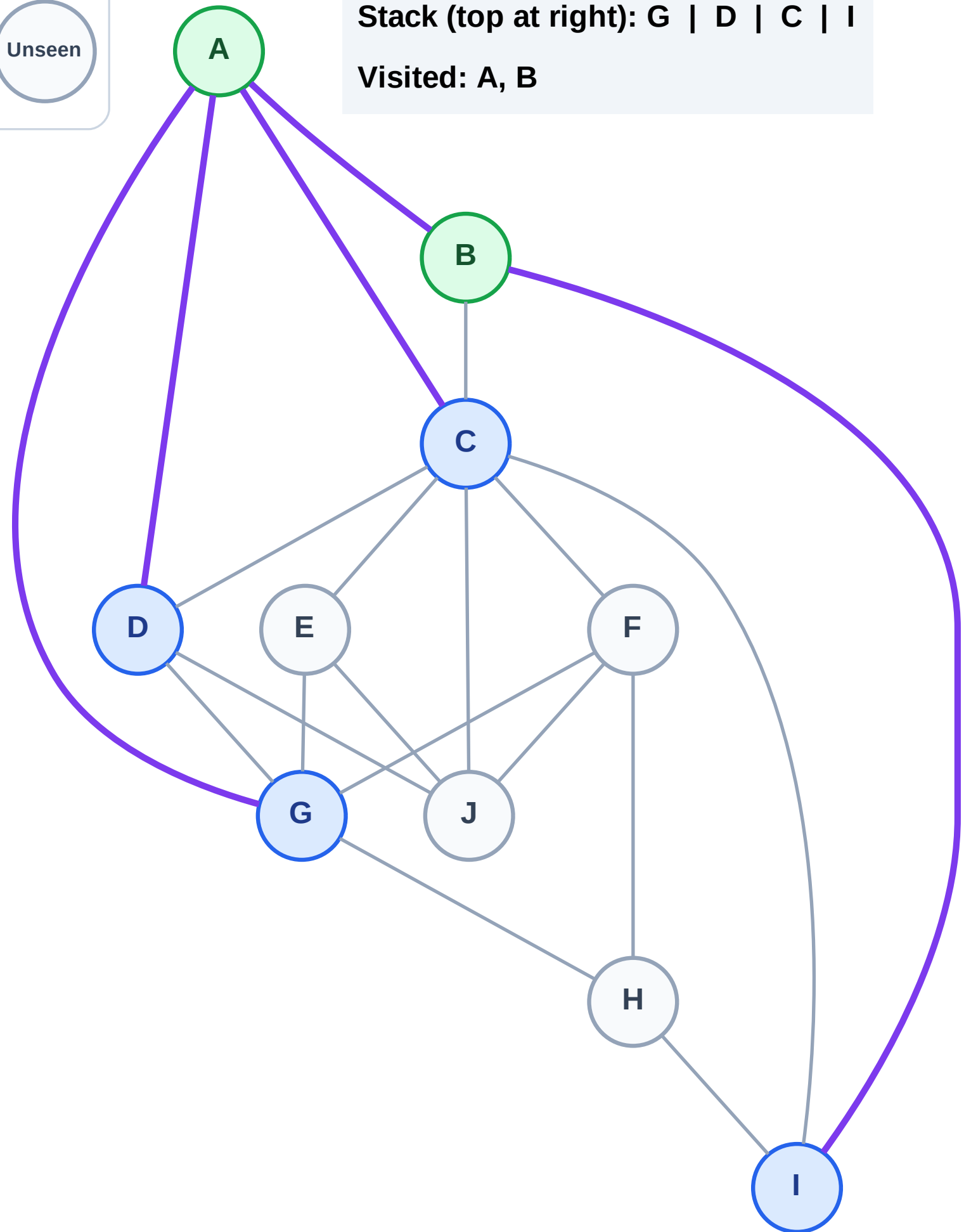
Step 5: Discover I from B

Legend



Stack (top at right): G | D | C | I

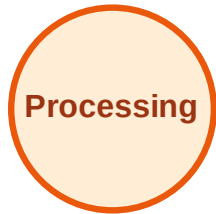
Visited: A, B



DFS Traversal

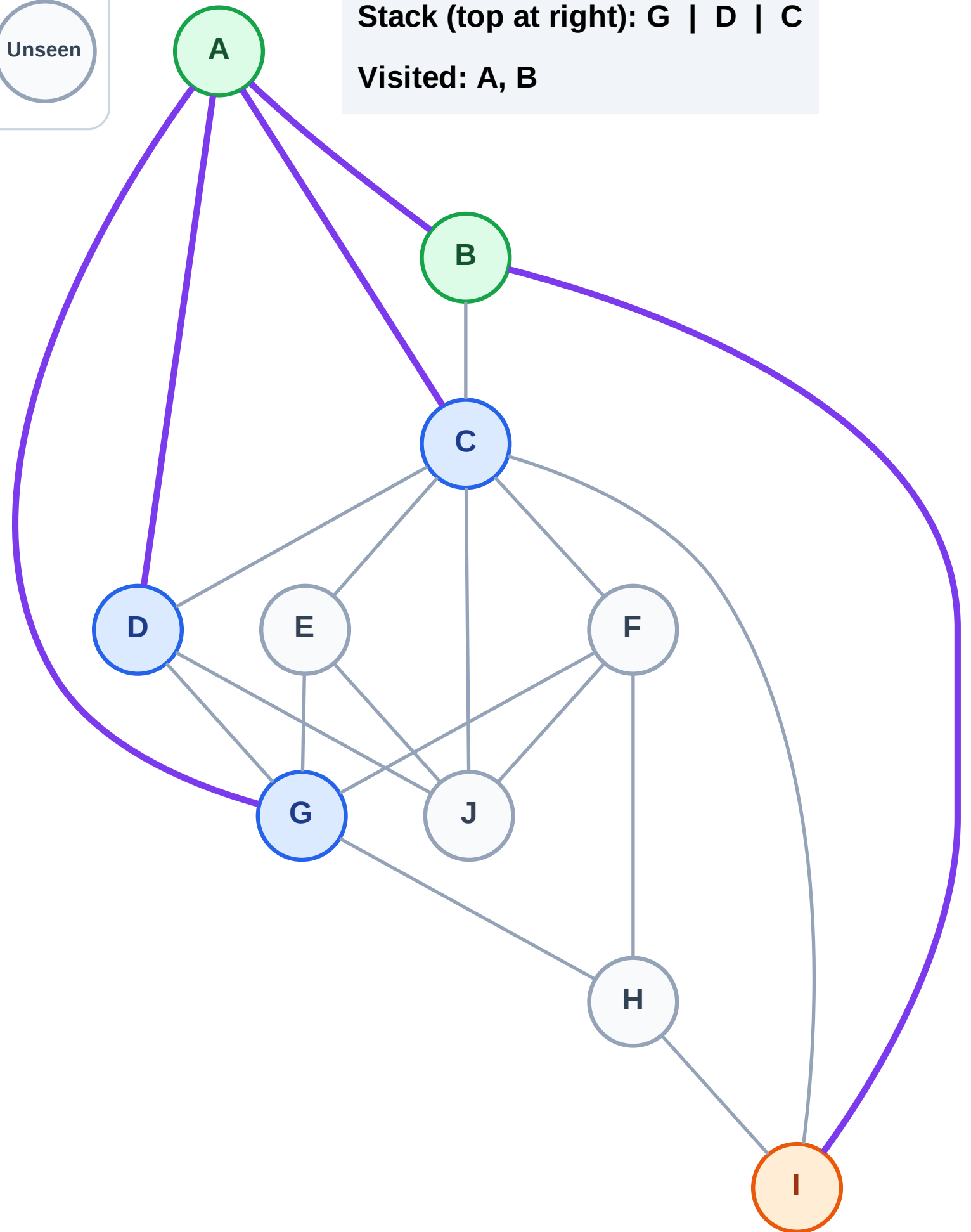
Step 6: Process node I

Legend



Stack (top at right): G | D | C

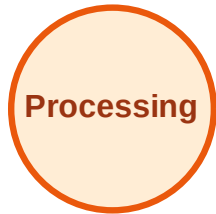
Visited: A, B



DFS Traversal

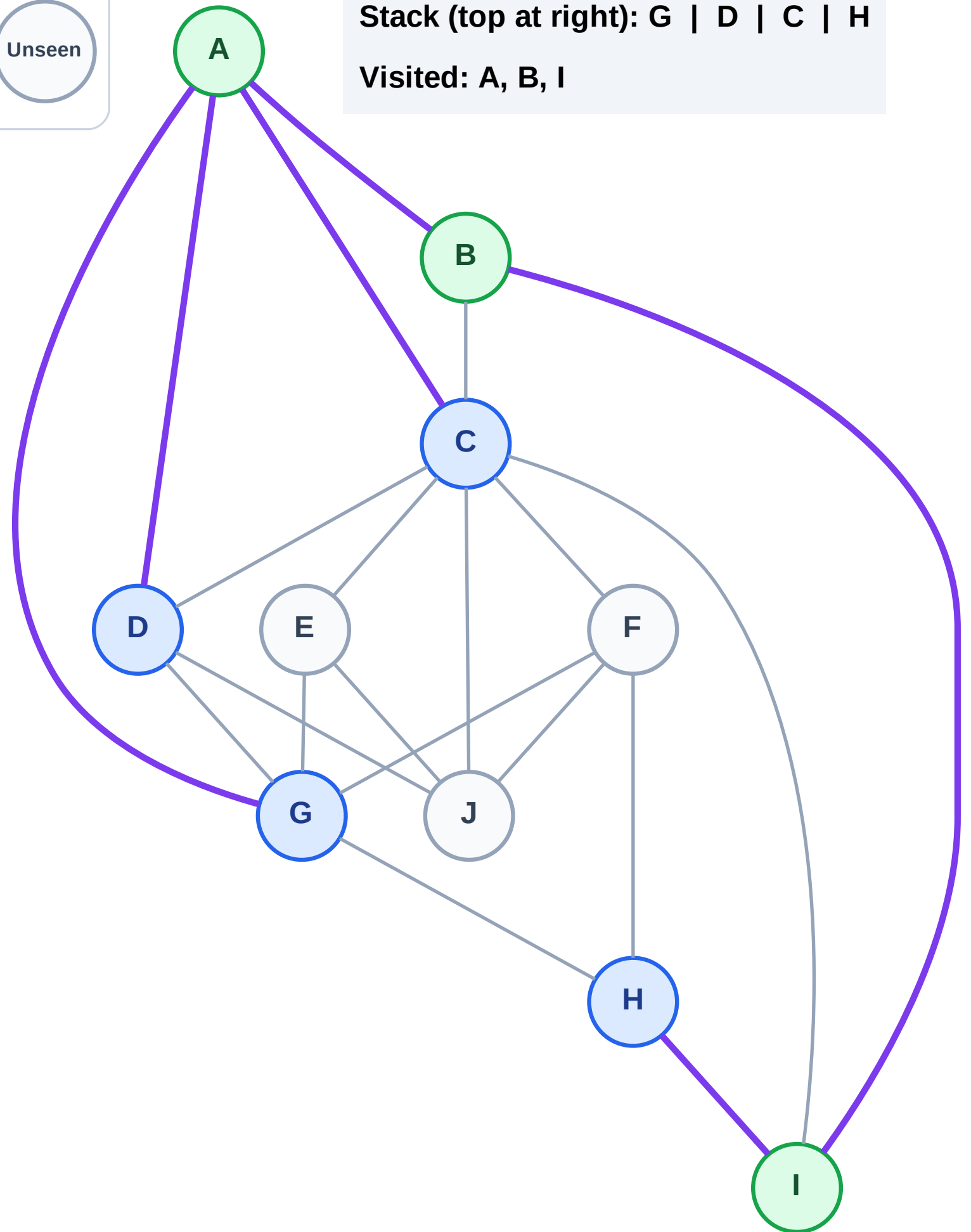
Step 7: Discover H from I

Legend



Stack (top at right): G | D | C | H

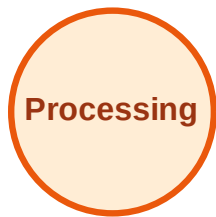
Visited: A, B, I



DFS Traversal

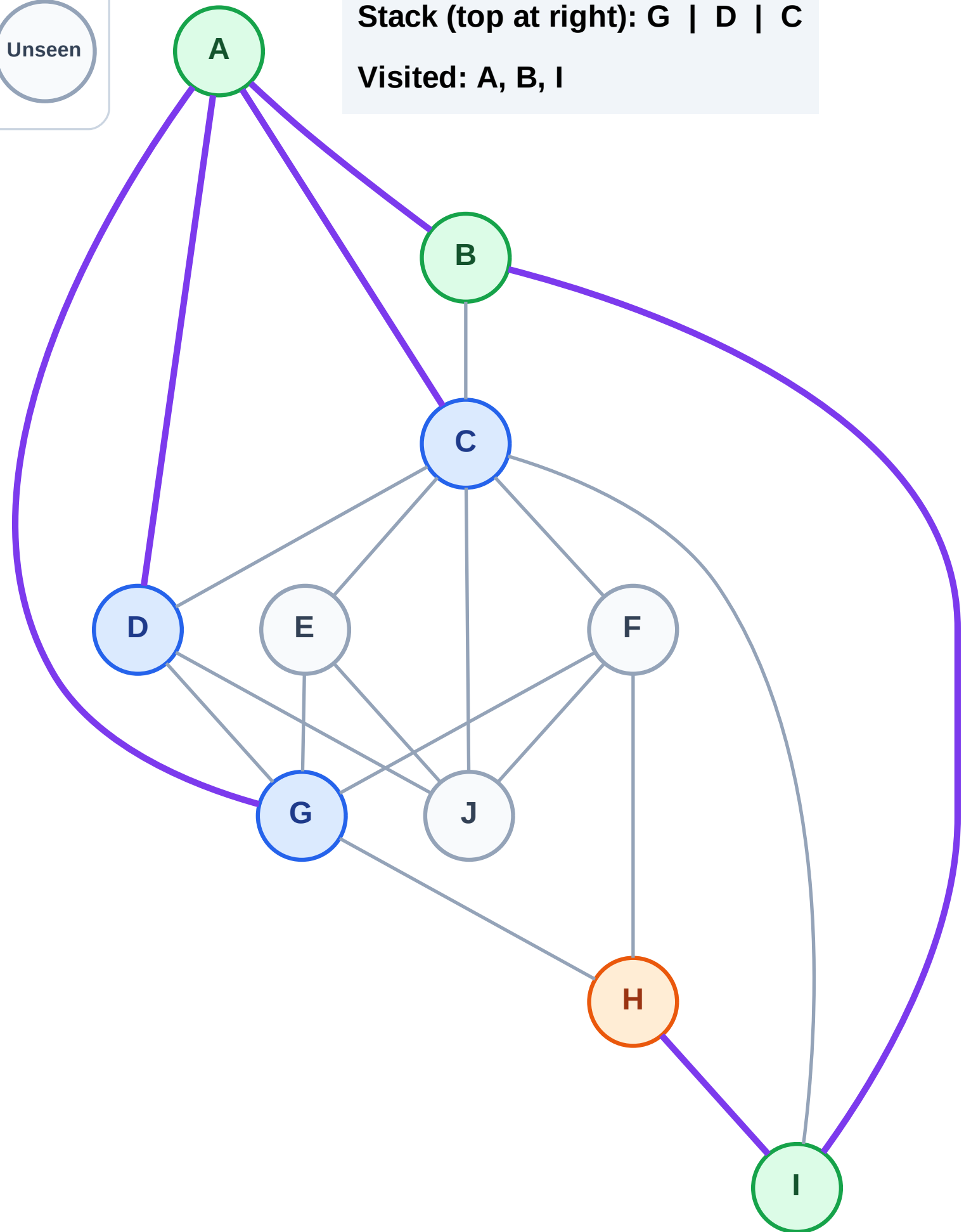
Step 8: Process node H

Legend



Stack (top at right): G | D | C

Visited: A, B, I



DFS Traversal

Step 9: Discover F from H

Legend

Complete

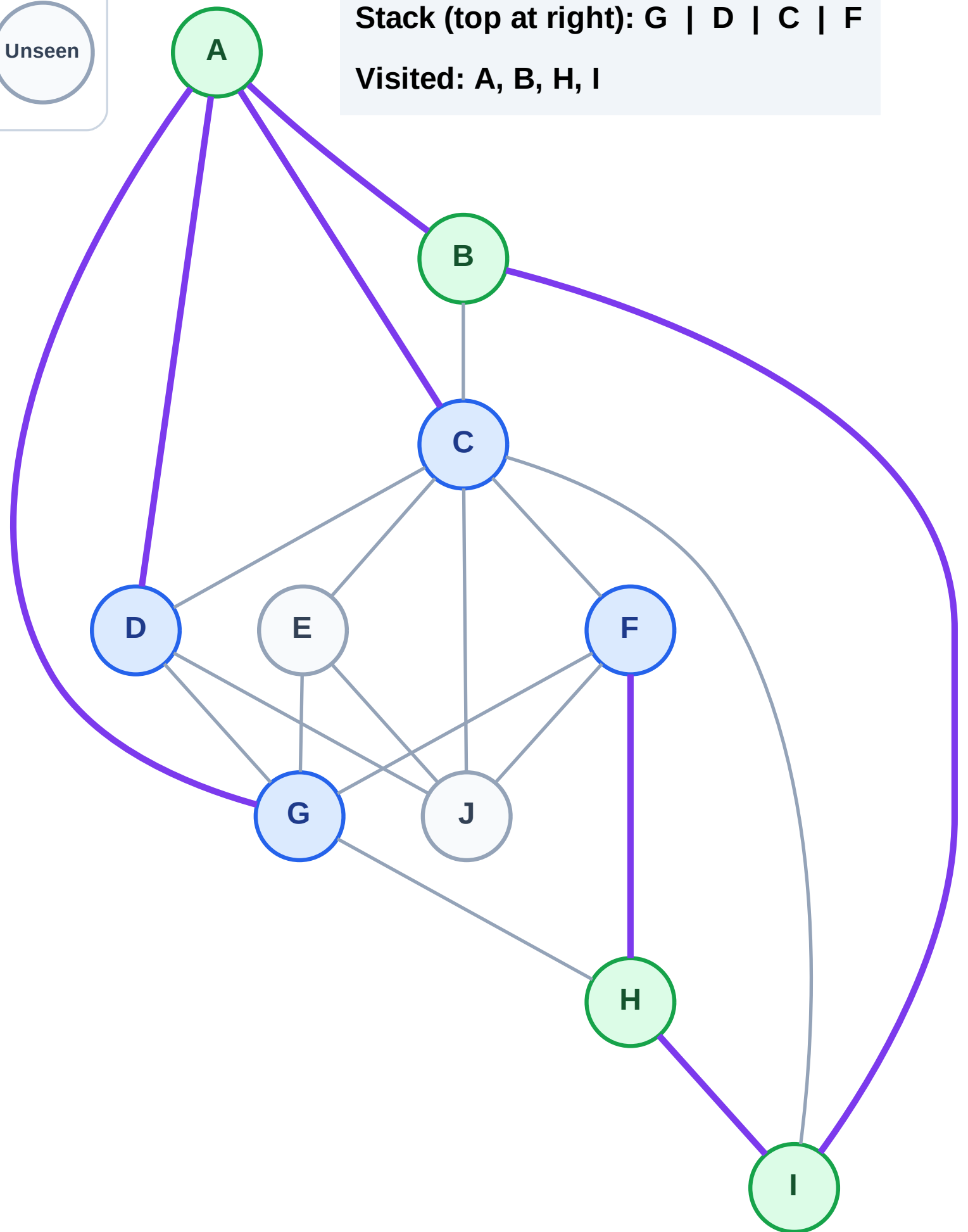
Processing

Discovered

Unseen

Stack (top at right): G | D | C | F

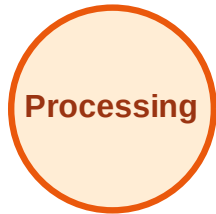
Visited: A, B, H, I



DFS Traversal

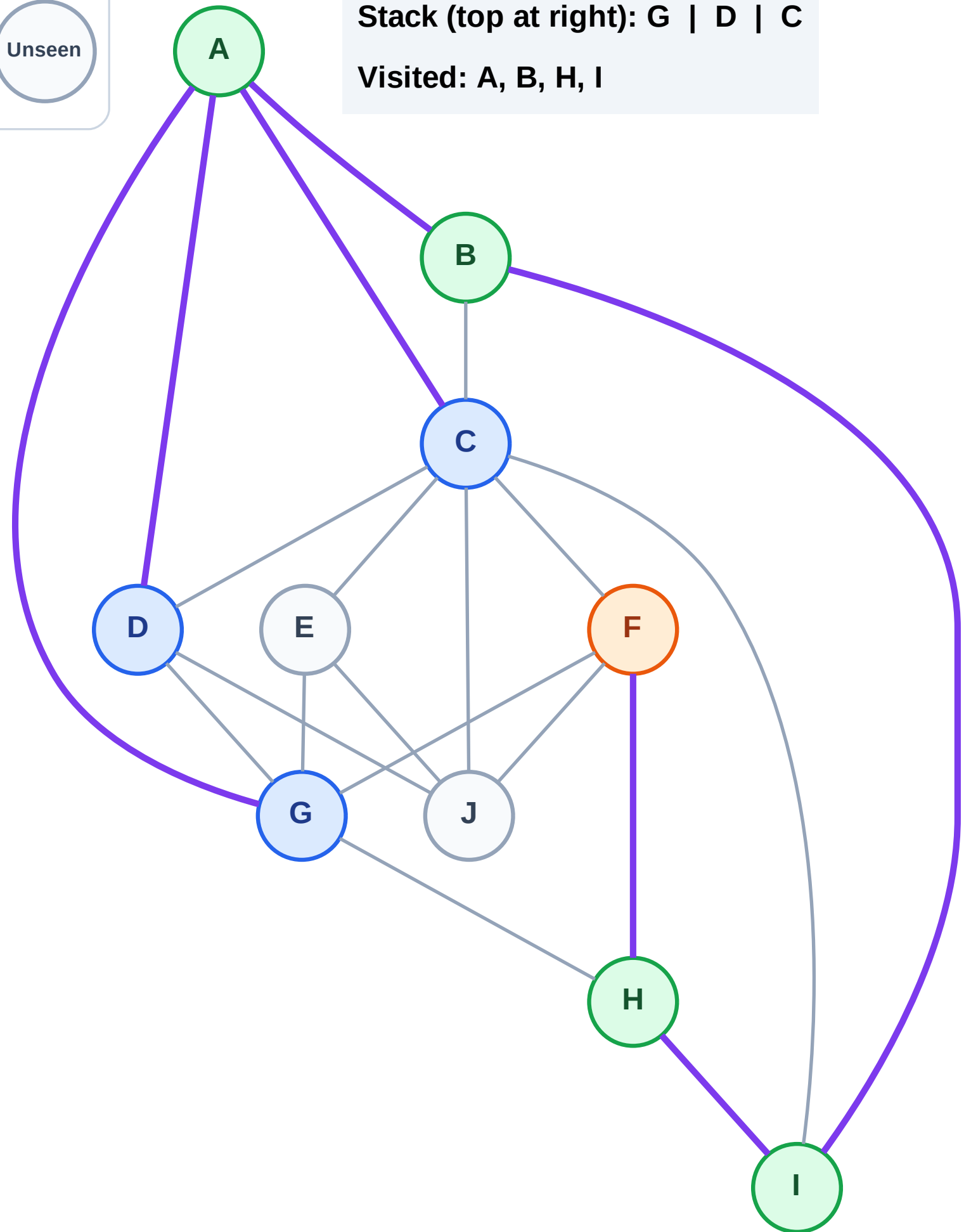
Step 10: Process node F

Legend



Stack (top at right): G | D | C

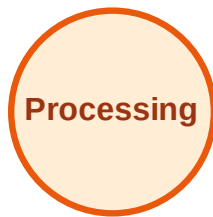
Visited: A, B, H, I



DFS Traversal

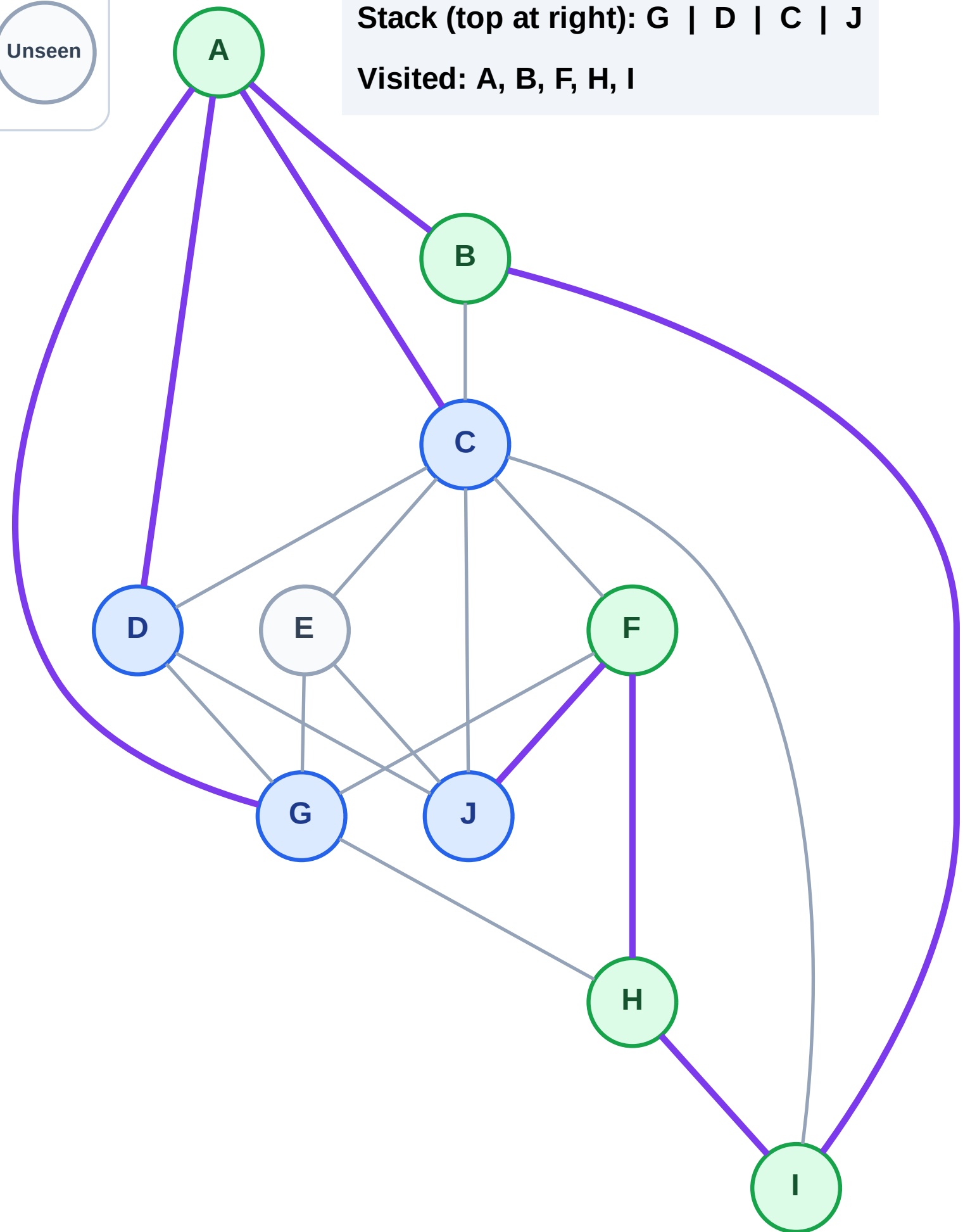
Step 11: Discover J from F

Legend



Stack (top at right): G | D | C | J

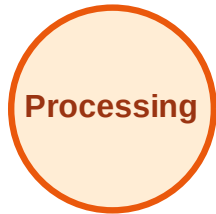
Visited: A, B, F, H, I



DFS Traversal

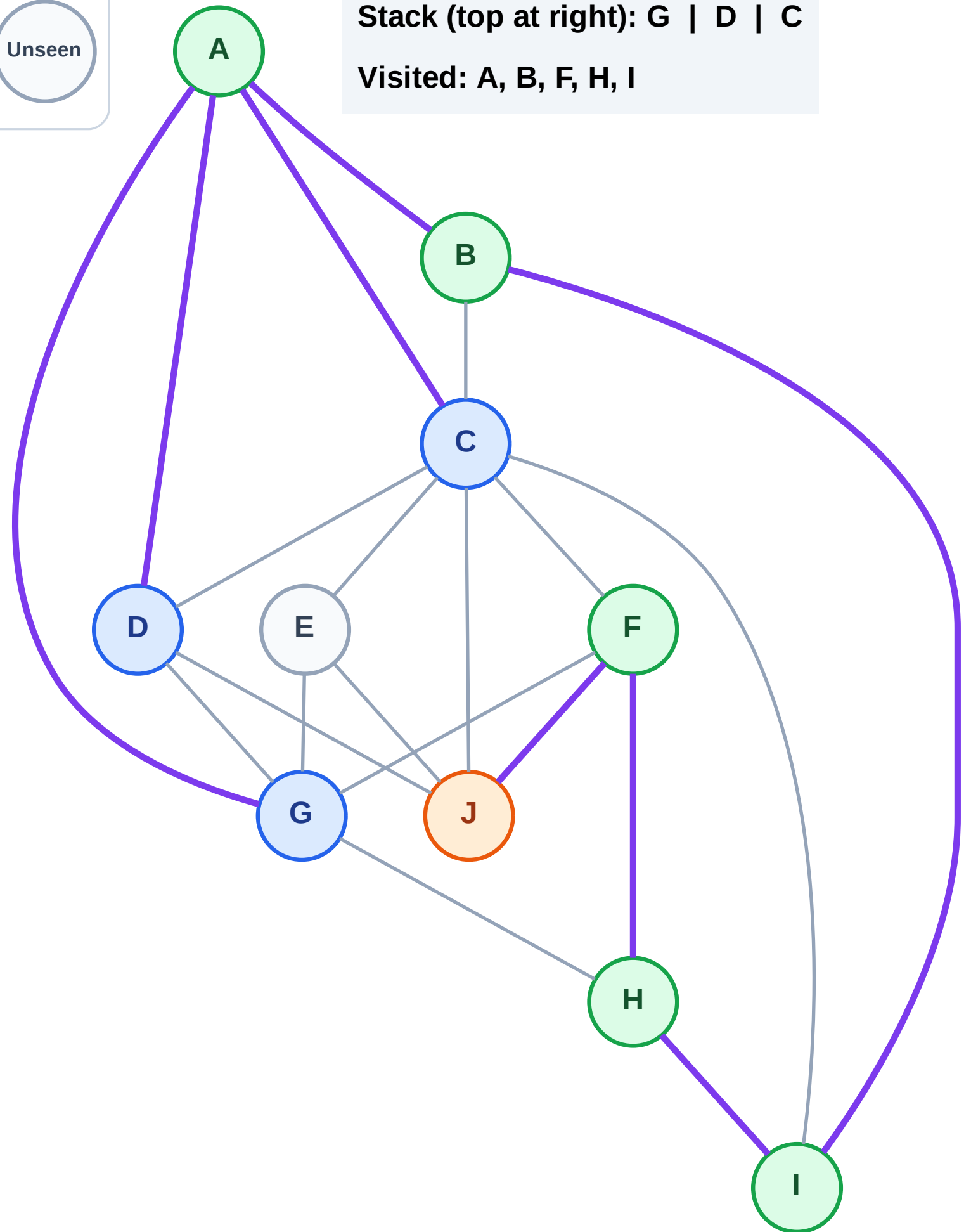
Step 12: Process node J

Legend



Stack (top at right): G | D | C

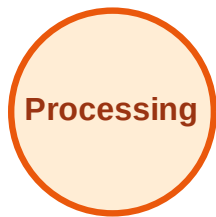
Visited: A, B, F, H, I



DFS Traversal

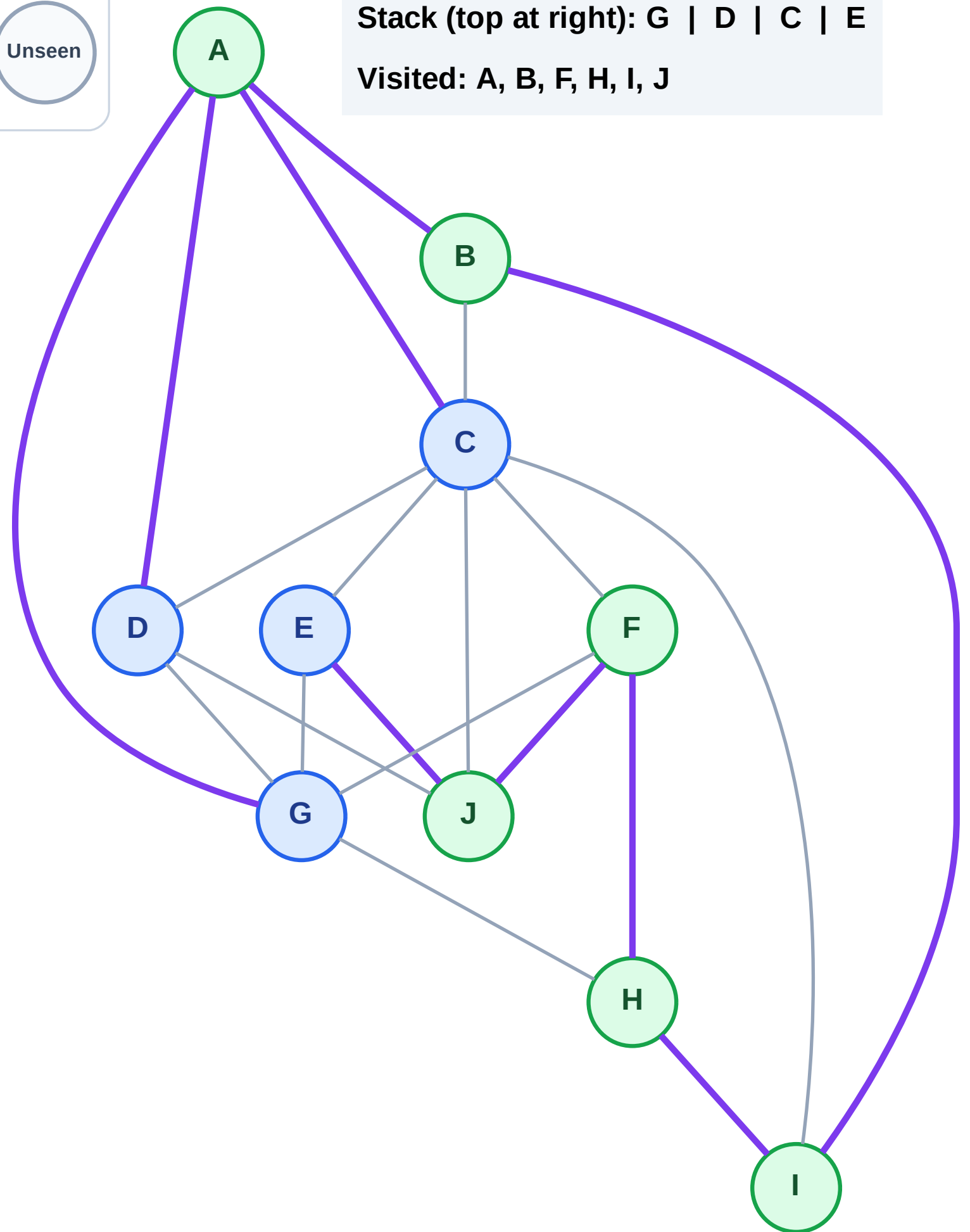
Step 13: Discover E from J

Legend



Stack (top at right): G | D | C | E

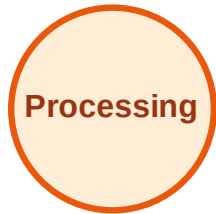
Visited: A, B, F, H, I, J



DFS Traversal

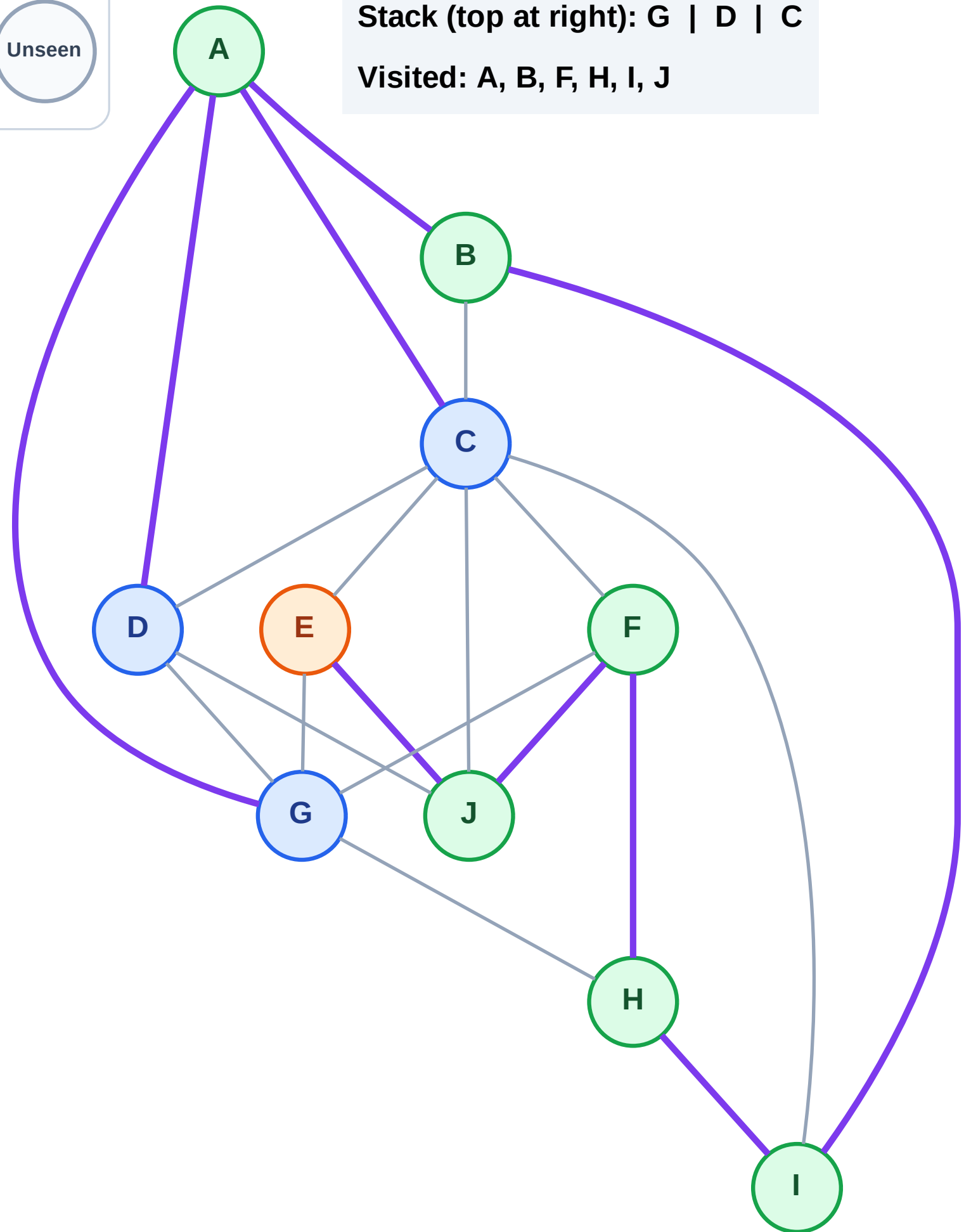
Step 14: Process node E

Legend



Stack (top at right): G | D | C

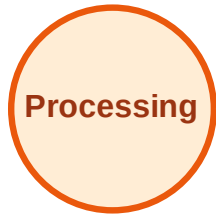
Visited: A, B, F, H, I, J



DFS Traversal

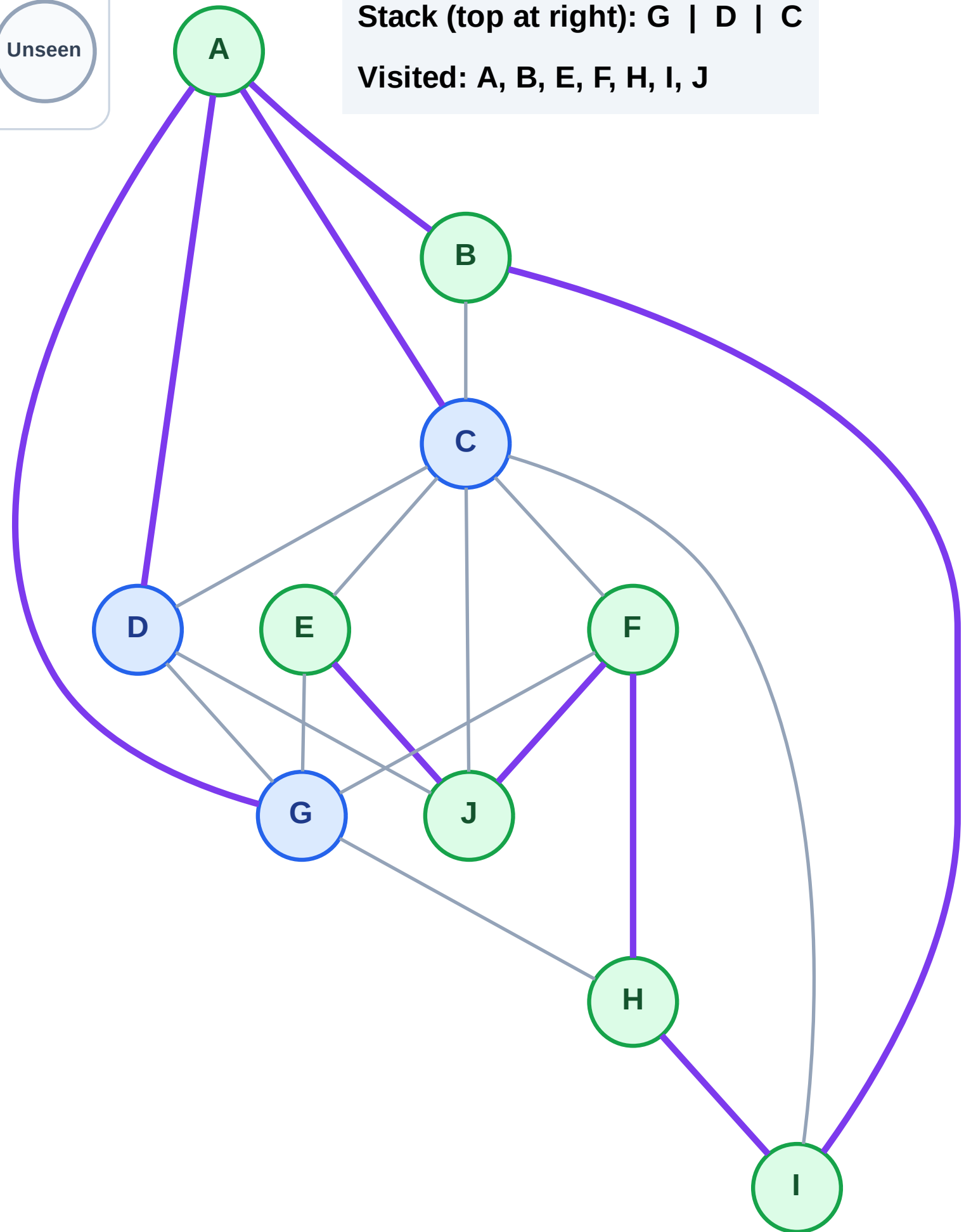
Step 15: No new neighbors from E

Legend



Stack (top at right): G | D | C

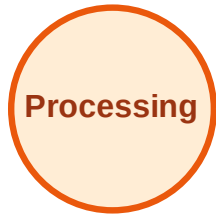
Visited: A, B, E, F, H, I, J



DFS Traversal

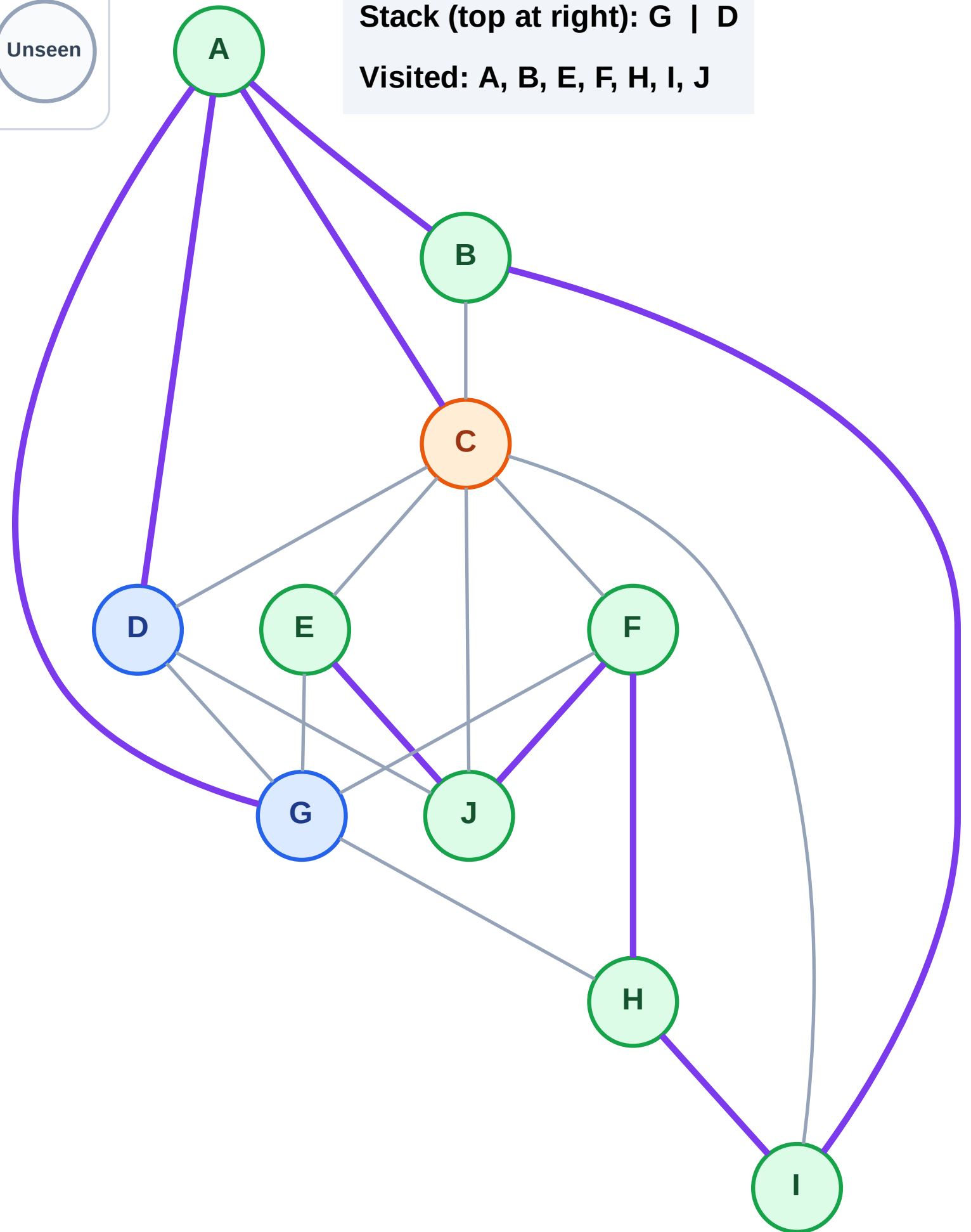
Step 16: Process node C

Legend



Stack (top at right): G | D

Visited: A, B, E, F, H, I, J



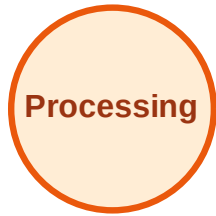
DFS Traversal

Step 17: No new neighbors from C

Legend



Complete



Processing



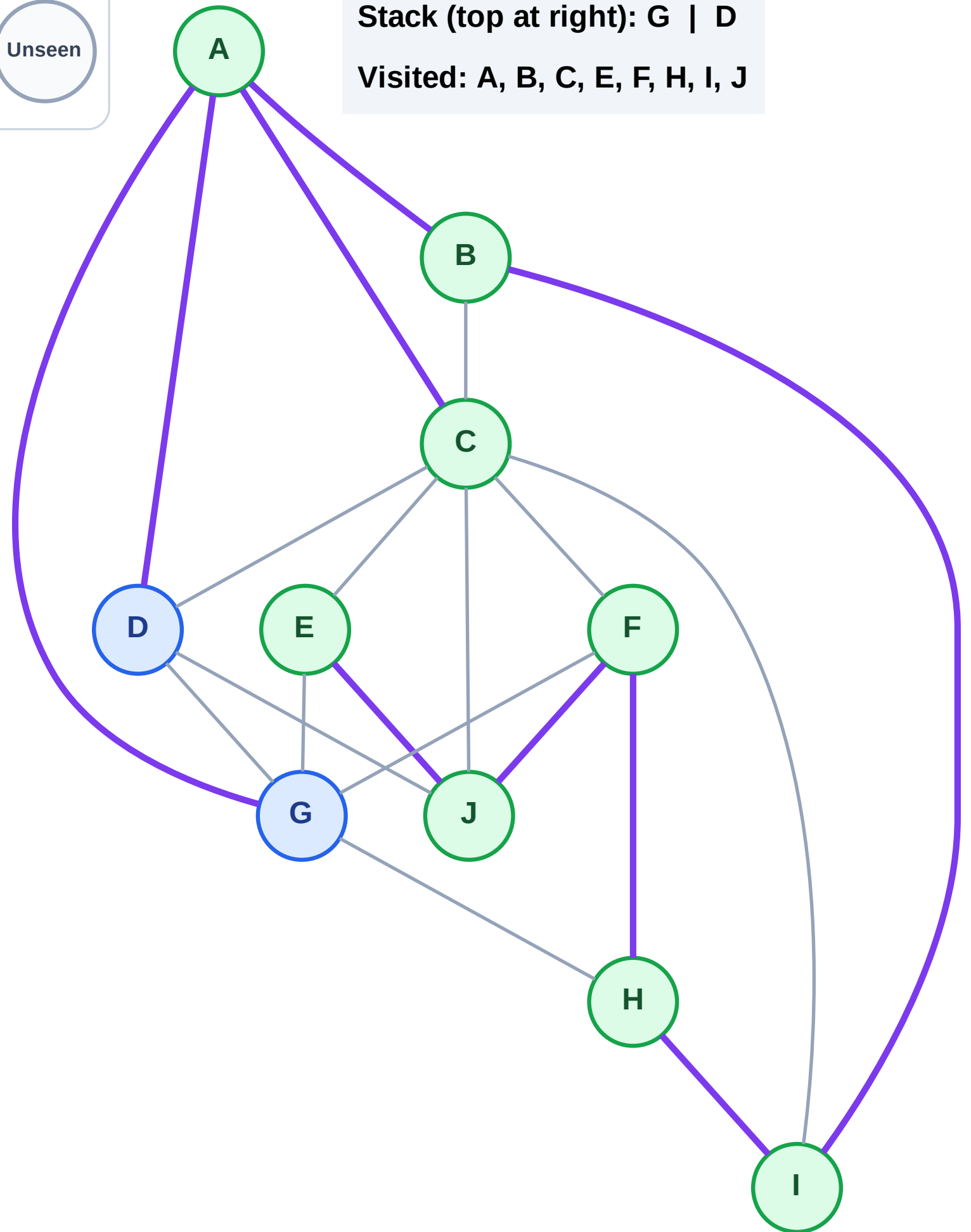
Discovered



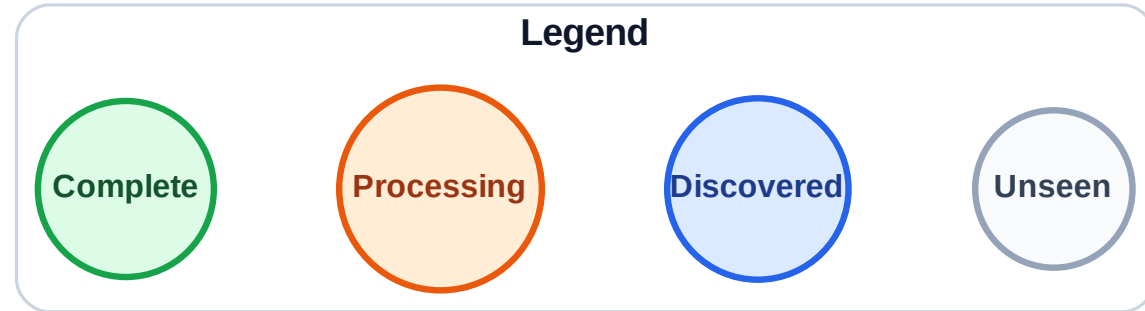
Unseen

Stack (top at right): G | D

Visited: A, B, C, E, F, H, I, J

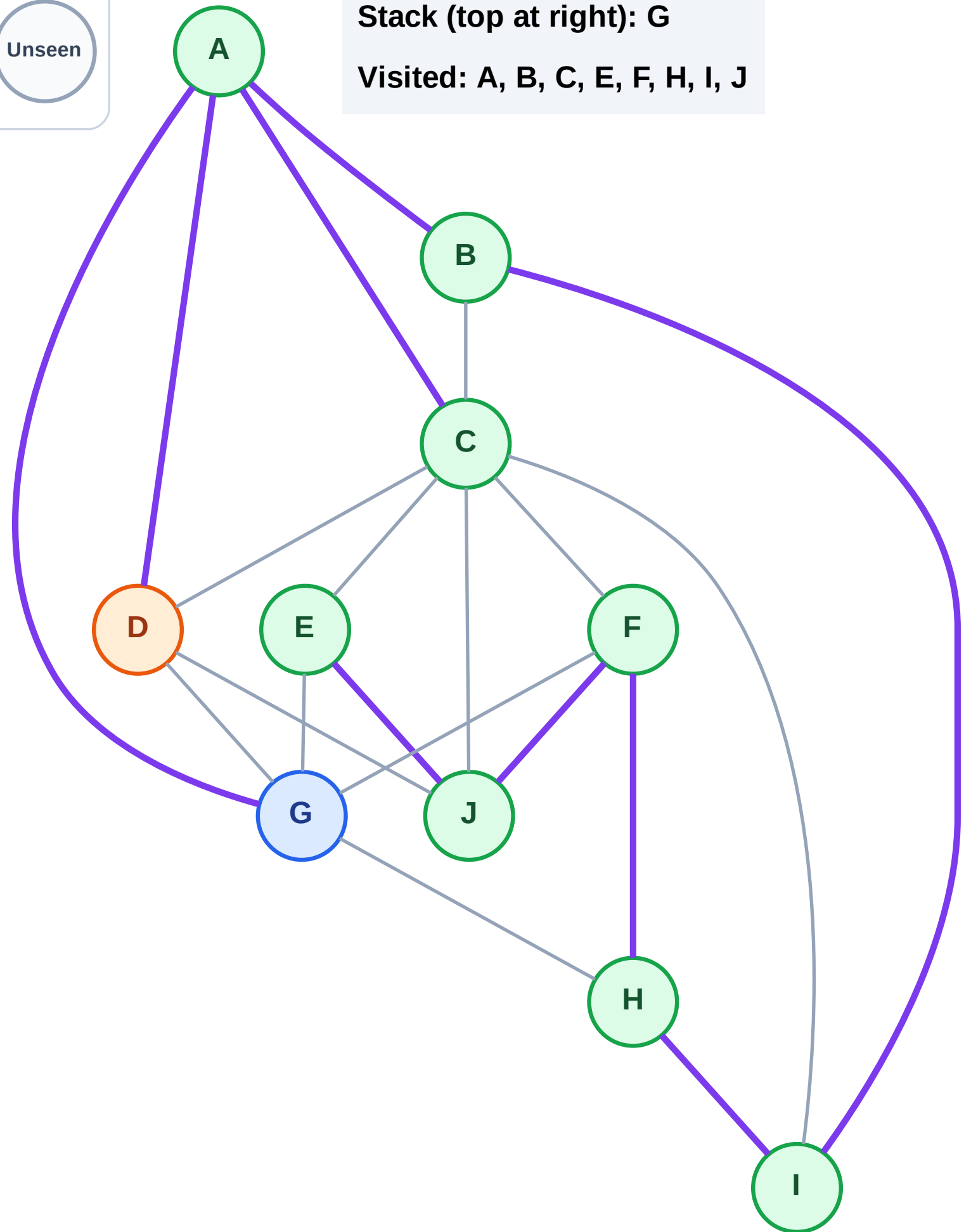


Step 18: Process node D



Stack (top at right): G

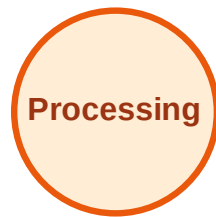
Visited: A, B, C, E, F, H, I, J



DFS Traversal

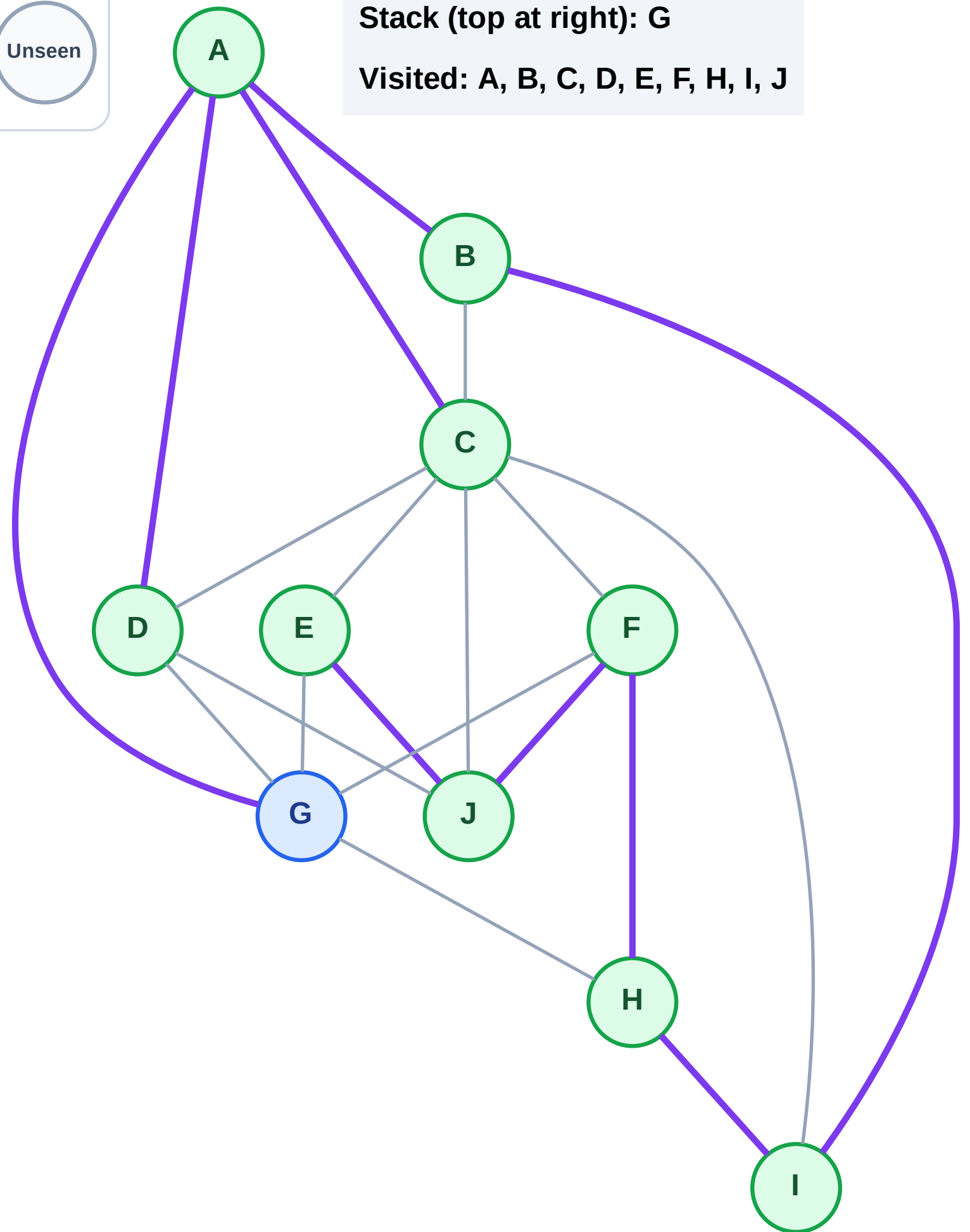
Step 19: No new neighbors from D

Legend



Stack (top at right): G

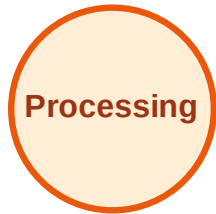
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

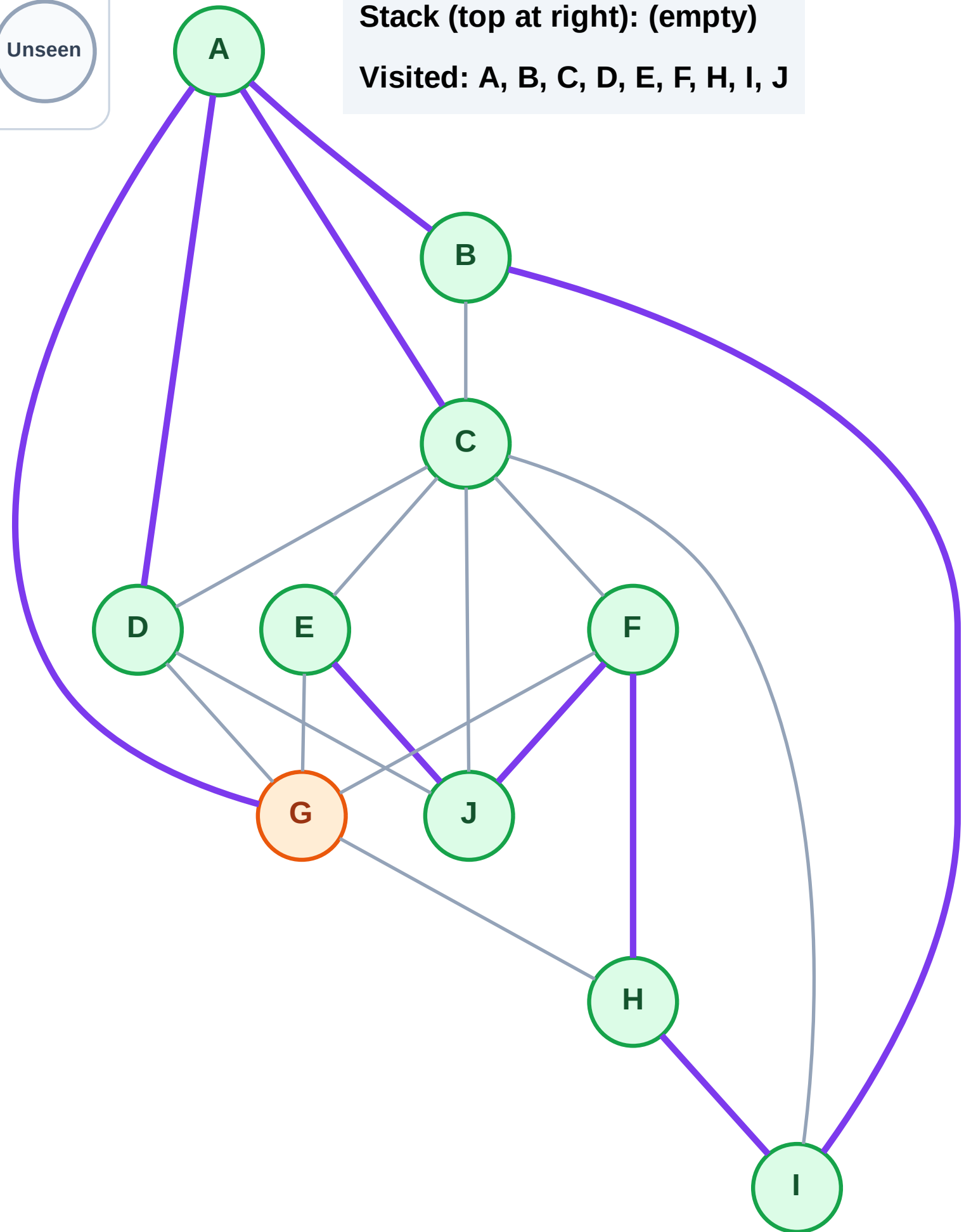
Step 20: Process node G

Legend



Stack (top at right): (empty)

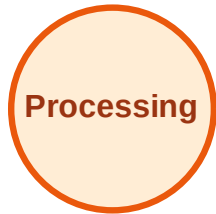
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 21: No new neighbors from G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

